A neural-mass mechanism for temporal-interference envelope demodulation: sigmoid curvature and near-Hopf amplification

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Abstract

Temporal-interference (TI) stimulation delivers a high-frequency carrier whose amplitude envelope beats at a low (e.g. alpha or gamma) frequency. An amplitude-modulated (AM) field carries no spectral power at its envelope frequency, so no linear element—passive membrane or synaptic kernel—can follow it; demodulation requires a nonlinearity, usually attributed to single-neuron ion channels. We propose and demonstrate a complementary, explicitly *network* mechanism at the mesoscopic scale whose two ingredients factorize. The population sigmoid *detects*: in a neural mass (a population of $\sim 10^4$ neurons) it is a square-law demodulator, $A_\Omega \simeq \frac{1}{2}\sigma''(v^*)\,\varepsilon^2 m$ (recovered envelope amplitude A_Ω; sigmoid curvature σ'' at operating point v^*; post-coupling polarization ε; modulation depth m), a curvature inherited from the single neuron's input–output nonlinearity and requiring no specific channel. The network *amplifies*: a second-order-synapse loop near a supercritical Hopf bifurcation resonantly boosts the recovered envelope at its natural frequency, the gain growing as $1/\gamma$ as the damping γ—the distance to the Hopf—vanishes. The two are separable: holding detection fixed, this gain is set by the network's proximity to criticality and is tuned by the coupling itself, and—being sharply frequency-selective—it amplifies the envelope-frequency response (population *timing*) far more than the mean firing rate. In a Jansen–Rit column an alpha-free carrier input yields a clean alpha output obeying the square law, carrier-independent at fixed polarization (the applied-field threshold rising with carrier frequency f_c) and sign-reversed by the curvature across the sigmoid inflection. The two-band laminar model (LaNMM) and the exact next-generation mean field (NMM2) confirm it—the latter with the demodulating nonlinearity *and* the amplifying coupling both derived rather than postulated. The mechanism is state-dependent through the operating point (PEIX) and connects TI to cross-frequency coupling and hierarchical amplitude modulation. The message: *wherever a fast-enough nonlinearity samples the carrier, the population sigmoid demodulates and the network amplifies*—the cortical column acting as a tuned AM receiver amplifier.

Keywords: temporal interference; transcranial stimulation; neural mass model; amplitude demodulation; Hopf bifurcation; cross-frequency coupling; Jansen–Rit; LaNMM.

Contents

1 Introduction

Temporal-interference stimulation [1] applies two sinusoids at f_1 and $f_2 = f_1 + \Delta f$ through separate electrode pairs (typically $f_1 \sim 1$–2 kHz, $\Delta f \sim 1$–100 Hz). Where the two fields overlap, their superposition is an AM signal whose carrier is near f_1 and whose envelope beats at the difference frequency Δf. The appeal is spatial: the envelope amplitude peaks in a focal, steerable region at depth, while superficial tissue sees only the (putatively inert) kHz carrier—a promise of non-invasive, focal, deep stimulation that explains the intense interest.

The difficulty is conceptual rather than technological. Writing the field as

$$s(t) = \varepsilon\,[\,1 + m\cos(\Omega t)\,]\cos(\omega_c t), \qquad \omega_c = 2\pi f_c, \quad \Omega = 2\pi\,\Delta f, \tag{1}$$

where m is the modulation depth, Δf the envelope (beat) frequency, f_c the carrier frequency, and ε the field amplitude (refined to the post-coupling polarization in §2.2). Its spectrum has support *only* at ω_c and $\omega_c \pm \Omega$ (Fig. 4): there is no energy at the envelope frequency Ω. A linear, time-invariant (LTI) system maps a sinusoid to a sinusoid of the same frequency; it can attenuate or phase-shift the three carrier lines but can *never* synthesize a new line at Ω. The membrane's passive RC low-pass—often invoked as "the neuron follows the slow envelope"—therefore cannot by itself demodulate TI. Demodulation requires a nonlinearity. This is precisely the founding lesson of AM radio [2]: a receiver cannot recover the message with linear components alone; it needs a nonlinear element (a square-law device or a diode) to shift spectral energy back to baseband, followed by a tuned filter to select the station and a low-pass to read the envelope.

Where, then, is the nonlinearity in the brain? The original passive-filtering intuition is now widely regarded as insufficient, since a linear filter cannot create the envelope line [3]. The leading biophysical account places the nonlinearity in single-neuron ion channels: Mirzakhalili et al. [3] show in Hodgkin–Huxley/Frankenhaeuser–Huxley axon models that TI works through nonlinear, ion-channel-mediated rectification, in which asymmetric voltage-gated conductances rectify the carrier and, after membrane low-pass, follow the envelope—the textbook rectify-then-filter detector. Other studies report that TI thresholds inherit the carrier's frequency dependence, suggesting a mechanism shared with direct kHz stimulation (conduction block, onset response) rather than a dedicated envelope channel [4, 5], and in some peripheral preparations the response is not driven by envelope extraction at all [5]. In vivo, TI reliably alters spike *timing* but not firing rate in the primate brain, and is much weaker than conventional low-frequency stimulation under human-relevant conditions [6]. Most tellingly, recent slice work finds that pyramidal and parvalbumin (PV) interneurons respond differently and that most pyramidal TI responses disappear when synaptic transmission is blocked, leading Caldas-Martinez et al. to conclude that TI is "largely a network phenomenon" [7]; computational network models point the same way [8].

We add a mechanism at the population (neural-mass) scale, from two ingredients already in standard cortical models: a static firing-rate *sigmoid*, whose nonzero curvature σ'' makes it a square-law demodulator, and a *second-order-synapse network* near a *Hopf bifurcation*, which both filters the carrier away and resonantly amplifies the recovered envelope at the network's natural frequency. Where existing accounts locate the nonlinearity in cellular excitability, ours is that the mesoscopic population transfer function already suffices. The sigmoid is not independent of single-cell physics—it is a coarse-grained summary of a population's input–output curves, thresholds, noise, and synaptic background—but once coarse-grained it is a *distinct mesoscopic nonlinearity* with enough curvature to detect envelopes, whatever microscopic mechanism supplies it. This is naturally consistent with the "network phenomenon" finding [7]: we give a minimal neural-mass implementation of such a population demodulator, in addition carrier-independent (at fixed polarization) and frequency-selective. We develop it first in a single Jansen–Rit column, analyzable in closed form, then in the laminar neural mass model (LaNMM), whose own alpha and gamma loops make the prediction concrete one step closer to cortex.

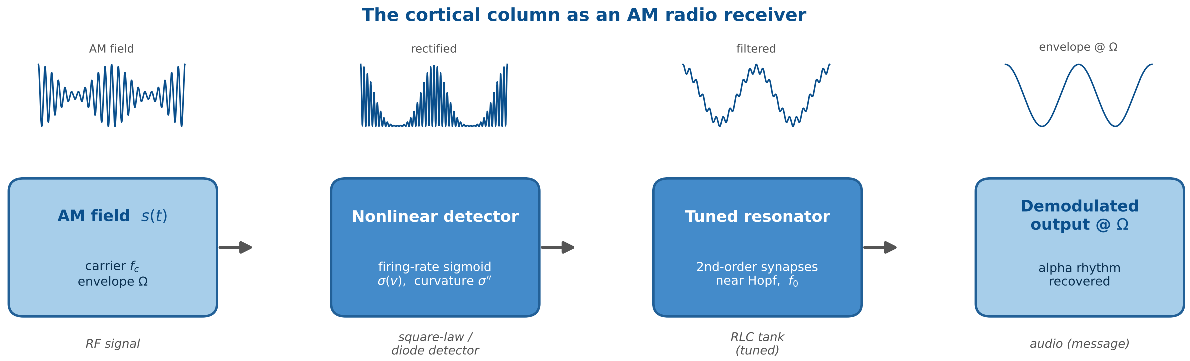

Figure 1: The cortical column as an AM radio receiver (schematic). An AM field (carrier f_c, envelope Ω) reaches a nonlinear detector (the firing-rate sigmoid, curvature σ''), whose output passes through a tuned resonator (the second-order-synapse network near a Hopf, natural frequency f_0), recovering the envelope as an oscillation at Ω. The stages map onto the RF front end, square-law/diode detector, RLC tank, and audio output of a classic AM receiver [2]. The firing-rate sigmoid plays the role of the detector; the second-order-synapse network near a Hopf is the tuned tank that selects and amplifies a particular envelope frequency, so TI "broadcasts" to whichever columns are tuned to its beat frequency.

This work builds on a line of research in which neural-mass models implement AM-radio principles *internally*, for the brain's own signals. In our cross-frequency-coupling account of predictive coding [9], the LaNMM realizes a columnar Comparator through sigmoid-enabled signal-to-envelope (SEC) and envelope-to-envelope (EEC) couplings, where error evaluation is literally demodulation—the sign-reversed prediction "un-modulates" the carrier. The Hierarchical Amplitude Modulation (HAM) framework [10] generalizes this to nested envelopes with constant-Q (log-spaced) bands, a candidate origin of $1/f$ spectra—again through the sigmoid's quadratic cross-term. The present work turns this machinery outward: if the cortex is an AM transceiver for its own rhythms, it can also be an AM receiver for an external carrier.

2 Theory

2.1 Neural mass models and the Jansen–Rit circuit

A neural mass model (NMM) collapses the activity of a large population ($\sim 10^4$) of similar neurons into a handful of population-averaged variables [11, 12]. Two elements recur in every such model. First, each synapse acts as a *linear filter* $\hat{K}$ that converts an incoming presynaptic firing rate $r(t)$ into a postsynaptic potential (PSP) $x(t) = \hat{K}[r]$; equivalently $r = \hat{L}[x]$ with $\hat{L} = \hat{K}^{-1}$ a differential operator whose few coefficients set the synapse's gain, decay time and delay [11]. In the Jansen–Rit (JR) family the kernel is the "alpha function" $h(t) = Aa\,t\,e^{-at}$ (excitatory; B, b for the inhibitory synapse), i.e. the second-order operator $\hat{L} = (Aa)^{-1}(\mathrm{d}/\mathrm{d}t + a)^2$, so each PSP obeys $\ddot{x} + 2a\dot{x} + a^2 x = Aa\,r(t)$. Second, the PSPs converging on a population sum to a mean membrane potential $v = \sum_s x_s$, which a static, saturating *transfer function*—the sigmoid—turns back into a population firing rate,

$$\sigma(v) = \frac{2e_0}{1 + e^{\rho(v_0 - v)}}, \tag{2}$$

bounded because firing rates are bounded [11]. The sigmoid is the only nonlinearity in the model, and—as we show—it is what makes (de)modulation possible. Its curvature, which does the demodulating, originates in the single neuron's own input–output nonlinearity: σ is the mean-field (ensemble) average of single-neuron transfer curves over a heterogeneous, noisy population, so σ'' is inherited from single cells and only reshaped—smoothed and broadened—by the spread of states,

Table 1: Jansen–Rit parameters [12].

Symbol	Meaning	Value
A, B	excit./inhib. PSP gain	3.25, 22 mV
a, b	synaptic rates	100, 50 s^{-1}
v_0, e_0, ρ	sigmoid inflection, max rate/2, slope	6 mV, 2.5 s^{-1}, 0.56 mV^{-1}
C	global connectivity	135
C_1, C_2, C_3, C_4	couplings	C, $0.8C$, $0.25C$, $0.25C$
p	external input (bifurcation knob)	0–400 Hz

thresholds and noise. It is not manufactured by the network; what the recurrent network adds is the resonant *amplification* (§2.3). Our claim is thus not that the nonlinearity is non-cellular, but that demodulation is read out at the population level by the generic sigmoid—whatever microscopic mix produces it, and without any specific channel.

This becomes exact one rung up the modeling ladder [11]. For a network of all-to-all coupled quadratic integrate-and-fire (QIF) neurons with Lorentzian-distributed excitabilities, the Ott–Antonsen ansatz yields a *closed, exact* mean field for the population firing rate r and mean potential v—the next-generation (Montbrió–Pazó–Roxin, MPR) mass [13], completed with the second-order synapses of neural-mass models in the exact "NMM2" theory [14, 15]:

$$\dot{r} = \frac{\Delta}{\pi} + 2rv, \qquad \dot{v} = v^2 + \bar{\eta} - \pi^2 r^2 + I(t), \tag{3}$$

with the applied current $I(t)$ entering linearly and the heterogeneity half-width Δ setting the smoothing. Here the demodulating nonlinearity is *derived, not postulated*: the quadratic v^2 term (with the rv coupling) is an exact square-law rectifier of the input, inherited directly from the single-neuron QIF dynamics. The Jansen–Rit sigmoid is the heuristic (NMM1) stand-in for this transfer—exact in the slow-synapse limit [15]—so our square-law result is a generic property of the mean field, not an artifact of expanding one postulated curve. The same exact theory already exhibits weak-field resonance and Arnold tongues that sharpen with self-coupling [15, 14], the near-Hopf amplification we exploit below.

The single-column JR circuit couples a pyramidal population to one excitatory and one inhibitory interneuron population through this synapse+sigmoid motif. Writing y_0 for the excitatory PSP driving the interneurons, y_1 and y_2 for the excitatory and inhibitory PSPs onto the pyramidal population, and the local-field-potential (LFP) proxy $v = y_1 - y_2$, with external input p,

$$\ddot{y}_0 = Aa\,\sigma\big(y_1 - y_2 + s(t)\big) - 2a\dot{y}_0 - a^2 y_0, \tag{4}$$

$$\ddot{y}_1 = Aa\left[\,p + C_2\,\sigma(C_1 y_0)\,\right] - 2a\dot{y}_1 - a^2 y_1, \tag{5}$$

$$\ddot{y}_2 = Bb\,C_4\,\sigma(C_3 y_0) - 2b\dot{y}_2 - b^2 y_2, \tag{6}$$

integrated as six first-order ODEs with the parameters of Table 1 [12]. The term $s(t)$ is the applied-field perturbation, introduced next; the read-out is $v = y_1 - y_2$.

For the biophysical realization (§4.6) we use a second model built from the same ingredients: the laminar neural mass model (LaNMM) [9, 16], which embeds this synapse+sigmoid motif in a two-band cortical column—a deep Jansen–Rit-like subcircuit (pyramidal P_1 with spiny-stellate (SS) and somatostatin (SST) interneurons; intrinsic alpha, ∼10 Hz) coupled to a superficial PING-like (pyramidal–interneuron network gamma) subcircuit (pyramidal P_2 with PV interneurons; intrinsic gamma, ∼40 Hz). The single JR column lets us isolate and analyze the mechanism; the LaNMM then exhibits it in a column with its own fast/slow loops and two native resonances. Everything below applies to both: the field couples through a sigmoid, and the recovered envelope drives a resonant circuit.

2.2 From applied field to a demodulated envelope

We model the applied TI field $E(t)$ (V/m) as a quasi-static perturbation of the membrane potential that the spike-generating nonlinearity reads. "Quasi-static" means we neglect electromagnetic propagation and induction (wavelengths $\gg$ head, frequencies far below tissue relaxation), so the field acts as an instantaneous polarization; taking it to add directly inside the sigmoid argument,

$$\boxed{\varphi(t) = \sigma\big(v(t) + \lambda\, E(t)\big), \qquad E(t) = E_0\,[\,1 + m\cos\Omega t\,]\cos\omega_c t} \tag{7}$$

This is the standard weak-field coupling $\delta V_m = \vec{\lambda}\cdot\vec{E}$ of transcranial-stimulation modeling [11], with λ a lumped polarization length. It is essential to keep three quantities distinct: the applied field $E(t)$, the induced membrane polarization $\delta V_m(t)$ *after* cable/membrane filtering, and the argument of the firing-rate map. The chain is

$$E(t) \xrightarrow[\text{cable/membrane}]{} \delta V_m(t) = \kappa(\mathbf{x}, \theta, f_c)\, E(t) \xrightarrow[\text{nonlinearity}]{} \sigma\big(v + \delta V_m\big) \xrightarrow[\text{network}]{} \text{resonance}, \tag{8}$$

with coupling $\kappa(f_c) = \lambda\,|H_m(f_c)|$ (mV per V/m) lumping cell geometry, orientation and the effective polarization length λ with the membrane band-pass H_m (§2.5). We therefore define ε as the *post-coupling polarization* amplitude that actually reaches the nonlinearity, $\varepsilon \equiv \kappa(f_c)E_0 = \lambda|H_m(f_c)|E_0$, with $s(t) \equiv \delta V_m(t)$, recovering Eq. (1)—not the raw applied field. In the main text we take the favorable quasi-static, direct limit $\kappa = \lambda$ ($|H_m| \to 1$), which isolates the sigmoid+resonance principle and is appropriate when a fast nonlinearity (axonal/nodal/terminal) samples the carrier before RC attenuation; §2.5 restores $H_m(f_c)$ and shows that the carrier dependence enters quadratically, $A_\Omega \propto |H_m(f_c)|^2$.

Expanding σ about an operating point v^* (a Volterra/Taylor expansion, as in [10]) gives $\sigma(v^* + s) \approx \sigma(v^*) + \sigma'(v^*)\, s + \frac{1}{2}\sigma''(v^*)\, s^2 + \ldots$ The linear term reproduces only the carrier sidebands. The quadratic term rectifies the AM field; keeping the baseband part of $s^2 = \varepsilon^2(1 + m\cos\Omega t)^2\cos^2\omega_c t$ (with $\cos^2\omega_c t \to \frac{1}{2}$),

$$\tfrac{1}{2}\sigma''\, s^2\big|_{\mathrm{LF}} = \tfrac{1}{4}\sigma''\varepsilon^2\Big[\big(1 + \tfrac{m^2}{2}\big) + 2m\cos\Omega t + \tfrac{m^2}{2}\cos 2\Omega t\Big], \tag{9}$$

so a line appears at exactly the envelope frequency Ω, with amplitude

$$\boxed{A_\Omega \simeq \tfrac{1}{2}\,\sigma''(v^*)\,\varepsilon^2\, m}. \tag{10}$$

In the language of [10], the quadratic cross-term $\kappa_2\, s_1 s_2$ between a slow and a fast input implements amplitude modulation/demodulation and the associated intermodulation lines. Equation (10) carries three predictions, all verified below: the recovered amplitude is quadratic in the drive ($A_\Omega \propto \varepsilon^2$, square-law detection); it follows the sigmoid curvature ($A_\Omega \propto \sigma''(v^*)$, hence zero at the inflection $v^* = v_0$); and, *at fixed post-coupling polarization* ε, it is independent of the carrier ω_c. This last prediction needs care: as a function of the *applied* field the response inherits the coupling $\varepsilon = \kappa(f_c)E_0$, so the field required for a given effect rises with f_c (§2.5)—consistent with the experimental consensus. Carrier independence is thus a statement about the recovered envelope at fixed polarization, not about the applied field.

2.3 Resonance, carrier independence, and the synaptic band-pass

Equation (10) contains no ω_c: at fixed post-coupling polarization ε the demodulated amplitude is independent of the carrier (as a function of the applied field it rolls off with f_c, §2.5). Under the direct coupling (7) the carrier need only be high enough that the synaptic low-pass suppresses it relative to the envelope, and "inert"—it must not itself fall in, or beat into, the network's responsive band, nor engage other carrier-specific mechanisms (the ion-channel/kHz pathway). Within those bounds the response is flat in f_c (Fig. 7, left). The synaptic suppression is concrete: each JR synapse applies the operator $\hat{L}_s$ with impulse response $h(t) = Aa\,t\,e^{-at}$, i.e. transfer function $H(\omega) = Aa/(i\omega + a)^2$, an over-damped second-order low-pass with corner $a/2\pi \approx 16$ Hz. It passes the $\sim$10 Hz envelope ($|H| \sim 0.7$) while suppressing a 100 Hz carrier to $\sim$2.5% (Fig. 7, right), so the carrier is filtered out of the loop *after* the sigmoid has used it to demodulate.

What turns this recovered envelope into a large, frequency-selective response is the network. Linearizing the mass about its fixed point gives eigenvalues $\lambda = -\gamma \pm i\omega_0$. Just below a supercritical Hopf, $\gamma \to 0^+$ and the mass is a weakly damped, high-Q resonator; the demodulated drive (10) at Ω excites it with gain

$$G(\Omega) \propto \left[(\omega_0^2 - \Omega^2)^2 + (2\gamma\Omega)^2\right]^{-1/2}, \qquad G(\omega_0) \propto \frac{1}{2\gamma\omega_0}, \tag{11}$$

so the steady response $\sim A_\Omega\, G(\Omega)$ is a Lorentzian peaked at $\Omega = \omega_0$ whose height diverges as the Hopf is approached. Just above the Hopf the mass is an autonomous limit cycle at ω_0, and the demodulated drive *entrains* it (an Arnold tongue), giving a response peak near ω_0 that grows with cycle amplitude.

2.4 Operating-point dependence

Because $A_\Omega \propto \sigma''(v^*)$, demodulation is governed by the curvature at the operating point, and in particular by how far v^* sits from the sigmoid inflection (Fig. 10). The Population Excitation–Inhibition Index (PEIX) of [9] is a related but distinct quantity: the sigmoid gain $\sigma'(v^*)$ normalized to its maximum $\sigma'(v_0) = e_0\rho/2$ at the inflection,

$$\mathrm{PEIX}(v^*) \equiv \frac{\sigma'(v^*)}{\sigma'(v_0)} = \mathrm{sech}^2\left[\tfrac{\rho}{2}\,(v^* - v_0)\right] \in (0, 1], \tag{12}$$

a dimensionless read-out of where v^* sits, the side given by $\mathrm{sign}(v^* - v_0)$. PEIX thus tracks σ' (maximal at the inflection v_0), whereas the demodulation gain tracks σ'' (zero at v_0, peaked on the flanks, and sign-reversing across it): the two are complementary, and PEIX peaks exactly where the demodulation gain vanishes. PEIX is therefore a convenient, measurable read-out of the operating point and hence an indirect predictor of σ'' and of TI efficacy. The mechanism predicts that anything shifting v^*—neuromodulation, attention, arousal, or pathology (psychedelics, PV interneuron dysfunction in Alzheimer's disease)—should scale, and can even flip the sign of, the TI response: a falsifiable handle.

2.5 Biophysical realization: carrier frequency, fast elements, and state

Carrier independence (at fixed polarization) holds for the direct coupling (7). Real somata are not direct: the membrane (and dendritic cable) low-passes the field before the spike nonlinearity. Modeling this as a first-order filter $H_m(\omega) = 1/(1 + i\omega\tau_m)$, the carrier amplitude reaching the detector is $\varepsilon_{\text{eff}}(f_c) = \varepsilon/\sqrt{1 + (\omega_c\tau_m)^2}$, and since $\Omega \ll \omega_c$ the three AM lines are attenuated almost equally, so

$$A_\Omega(f_c) \;\simeq\; \tfrac{1}{2}\,\sigma''(v^*)\, m\, \varepsilon_{\text{eff}}^2 \;=\; \frac{A_\Omega(0)}{1 + (\omega_c\tau_m)^2} \;\xrightarrow[\omega_c\tau_m \gg 1]{}\; A_\Omega(0)\,\frac{1}{(\omega_c\tau_m)^2}. \tag{13}$$

The demodulated response rolls off as $1/f_c^2$ above the membrane corner (Fig. 8). For a pyramidal soma/dendrite the passive time constant is $\tau_m \sim$10–30 ms (human L2/3 $\sim$12–22 ms, mean $\sim$16 ms; corner $\sim$10 Hz), so the carrier-power factor $|H_m|^2$ is only $\sim 10^{-4}$ at 1 kHz, $\sim 2.5 \times 10^{-5}$ at 2 kHz and $\sim 10^{-6}$ at 10 kHz. Passive somatic demodulation is therefore negligible at TI's kHz carriers—a four- to six-order-of-magnitude suppression.

The escape is that the demodulating nonlinearity need not sit at the soma. At a fast element (axon, axon initial segment (AIS), node of Ranvier; $\tau \approx 0.2$ ms, effective corner $\sim$800 Hz) the carrier is only mildly attenuated in the kHz band ($|H|^2 \approx 0.39$ at 1 kHz, 0.14 at 2 kHz, still $\sim 6 \times 10^{-3}$ at 10 kHz; Fig. 8). A square-law nonlinearity there produces a real envelope-frequency current, and that low-frequency component is *not* attenuated by the membrane—it is exactly what the slow network reads. The per-neuron modulation is subtle, but two factors make it count: the envelope sits in the band the membrane passes, and the population near a Hopf amplifies it by $G \sim 1/\gamma$ (Eq. (11)). Subtle demodulation at fast elements, resonantly amplified at the population, is thus a plausible route for TI at realistic carriers—consistent with ion-channel rectification at the node [3] but adding network resonance; our direct-coupling demonstration is the $\tau \to 0$ idealization of this limit, and a direct kHz-carrier run (Fig. 9) confirms it.

Because transcranial electrical stimulation (tES)—and TI, a tES with a kHz carrier—is weak and *modulatory*, the fast-element route is a small, sustained bias of the most excitable compartments, not "spiking the AIS at the beat frequency." Since the applied field adds to endogenous drive, $V(t) = V_0 + \delta V_{\mathrm{tES}}(t) + \eta(t)$, there is no strict minimum field: near threshold a tiny polarization biases spike probability and timing through stochastic resonance, whose inverted-U signature tES and transcranial random-noise stimulation (tRNS) display [17, 18, 19, 20]. This maps onto the neural mass, whose sigmoid (2) is itself a mean-field stochastic-resonance device—its smooth slope arises from the spread of thresholds and noise across the population—so a weak bias shifts the mean rate through σ (gain σ') while the demodulated envelope emerges through the curvature σ''.

Everything here is strongly state-dependent: the demodulation gain scales with $\sigma''(v^*)$ and the amplification with $1/\gamma$, both set by neuromodulation, attention, arousal and pathology (PEIX), so TI and transcranial magnetic stimulation (TMS) efficacy should vary with brain state rather than being fixed by the stimulus. The 100 Hz carrier used throughout is not merely convenient: there the membrane attenuation is mild for any moderately fast element ($|H_m|^2 \approx 0.98$ at $\tau = 0.2$ ms, ~ 0.1 at $\tau = 5$ ms, the fast-spiking-interneuron range), so demodulation does not need the fastest nodal channels, and a protocol with $f_c \sim 100$ Hz and Δf swept through a region's resonant band is a clean near-term test. The trade-off is spatial—a 100 Hz carrier is less inert outside the interference zone than a kHz one (kHz carriers being subthreshold there, the source of TI's focality)—so 100 Hz probes the mechanism most cleanly while kHz optimizes selectivity.

3 Methods

We integrate the single-column JR model of §2.1 (Eqs. (4), parameters in Table 1) as six first-order ODEs. The TI field enters only the pyramidal output sigmoid, Eq. (4), as $s(t)$; the read-out $v = y_1 - y_2$ is *not* fed the field directly, so any power at Ω in v is produced by the nonlinearity and the recurrent loop, not by trivial pass-through. Unless stated otherwise $f_c = 100$ Hz and $m = 1$.

We integrate with a fixed-step fourth-order Runge–Kutta scheme, $\Delta t = 2 \times 10^{-4}$ s (50 samples per carrier cycle at 100 Hz; reduced to 10^{-4} s for carrier sweeps up to 300 Hz). For each measurement the system is settled for 10–16 s (longer near the Hopf, where damping is weak) before a 3–20 s measurement window; the resonance sweeps use a 20 s window so the lock-in averages the limit-cycle beating on the entrainment side. Parameter sweeps are vectorized: the state is an $(N, 6)$ array advanced simultaneously for all N parameter combinations, so a full resonance map runs in seconds. The bifurcation structure (Fig. 2) is located by solving, for each p, the self-consistent scalar equation for the resting fixed point $v = y_1 - y_2$, forming the 6×6 Jacobian numerically, and diagonalizing it; the supercritical Hopf is the p at which the leading eigenvalue pair crosses the imaginary axis with $\mathrm{Im}\,\lambda$ in the alpha band. This gives $p \approx 315$ Hz and $f_0 = \mathrm{Im}\,\lambda/2\pi \approx 11.1$ Hz, in line with the JR bifurcation analysis of [21]; for $p > 315$ the fixed point is a stable focus (resonator) and for $p < 315$ an alpha limit cycle.

The response at the envelope frequency is read out with a software lock-in amplifier referenced to Ω. We multiply $v(t)$ by quadrature references and integrate over a Hann-windowed measurement

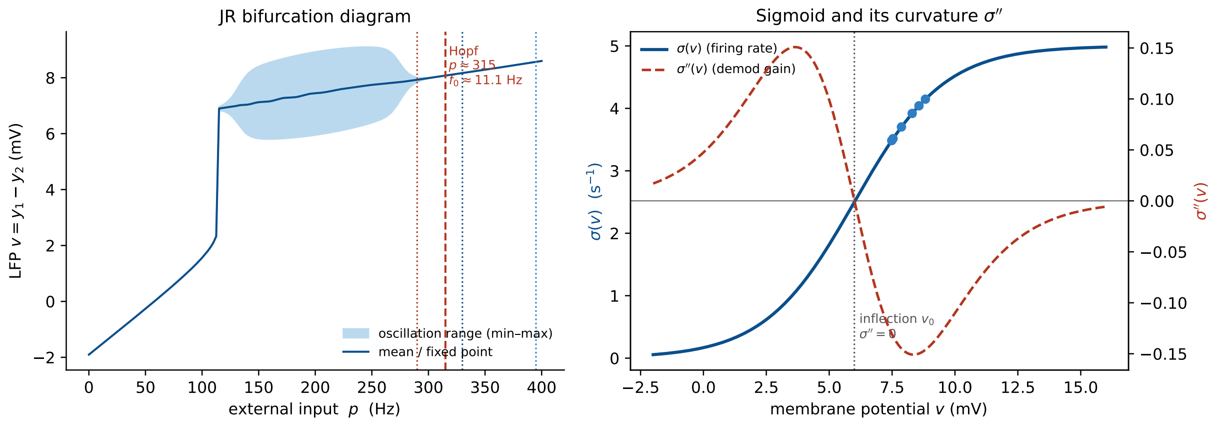

Figure 2: The Jansen–Rit operating regime: bifurcation and sigmoid curvature. *Left:* JR bifurcation diagram—LFP range vs. external input p. A limit cycle (alpha) exists between a saddle-node-on-invariant-circle (SNIC) bifurcation near $p \approx 115$ and the supercritical Hopf at $p \approx 315$; above the Hopf the fixed point is a stable focus, $f_0 \approx 11.1$ Hz. *Right:* the sigmoid $\sigma(v)$ and its curvature $\sigma''(v)$, which sets the demodulation gain and vanishes at the inflection v_0. Dots mark the operating points reached by the closed-loop model for the p values used here.

window to obtain the in-phase (I) and quadrature (Q) components,

$$I = \frac{2}{\sum_k w_k} \sum_k w_k \, (v_k - \bar{v}) \cos \Omega t_k, \quad Q = \frac{2}{\sum_k w_k} \sum_k w_k \, (v_k - \bar{v}) \sin \Omega t_k, \quad A_\Omega = \sqrt{I^2 + Q^2}, \tag{14}$$

with w a Hann window and $\bar{v} = \sum_k w_k v_k / \sum_k w_k$ the windowed mean. For a pure tone $v = A\cos(\Omega t + \phi)$ the product-to-sum identity gives $I \to A\cos\phi$ and $Q \to A\sin\phi$, so $A_\Omega \to A$ exactly—the factor $2/\sum_k w_k$ is the normalization that returns the physical amplitude, and A_Ω equals twice the magnitude of the windowed Fourier coefficient at Ω. The one subtlety is that out-of-band components must not leak into the estimate; the dangerous one is the DC level of v ($\bar{v} \sim 8$ mV, far larger than the demodulated signal). Over an exact integer number of Ω-periods $\sum_k w_k \cos \Omega t_k = 0$ and there is no leakage, but in general it is nonzero, so we subtract $\bar{v}$ explicitly; this makes the metric exact for a tone and robust down to the 10^{-8} level needed for the kHz analysis (§2.5). The signed in-phase component I alone recovers $\frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ including its sign, which we use to verify the curvature law (Fig. 10).

Three auxiliary constructions support the main results. An *open-loop detector* passes the AM field through σ at a fixed operating point v^* (no feedback) and applies Eq. (14), isolating the pure demodulation law (Figs. 7, 10, 8). The *bifurcation diagram* records the field-free min/max of v vs. p, each run initialized at its fixed point plus a small kick. The *two-dimensional map* evaluates A_Ω over an (Ω, p) grid. We verify the mechanism three ways: the square law (A_Ω vs. ε on a well-damped focus, expecting a log–log slope of 2); a linearization control (replacing σ in Eq. (4) by its tangent at v^*, which must abolish the Ω response); and a check of σ', σ'' against finite differences. All code is provided (`jr_demod.py`, `jr_analysis.py`, `make_figures.py`, `analyses_v2.py`, `figures_v2.py`, `khz_analysis.py`; see `README_Code.md`).

We complement the single-column analysis with simulations of the laminar neural mass model (LaNMM) of §2.1. We integrate the full 28-variable column [9, 16], driving a single population with an AM input

$$e(t) = I_0 + A\,[\,1 + \cos \Omega t\,] \cos(\omega_c t + \phi), \tag{15}$$

the LaNMM counterpart of Eq. (1) (modulation amplitude A, beat $\Omega = 2\pi\Delta f$, carrier $\omega_c = 2\pi f_c$), with the remaining drives held flat (Fig. 3). Integration uses an adaptive RK45 scheme (2 ms record step). In place of the lock-in (Eq. (14)) we read the alpha-band (8–12 Hz) power of the P_1 and P_2 membrane potentials from the Hilbert envelope over a post-transient window—a phase-robust

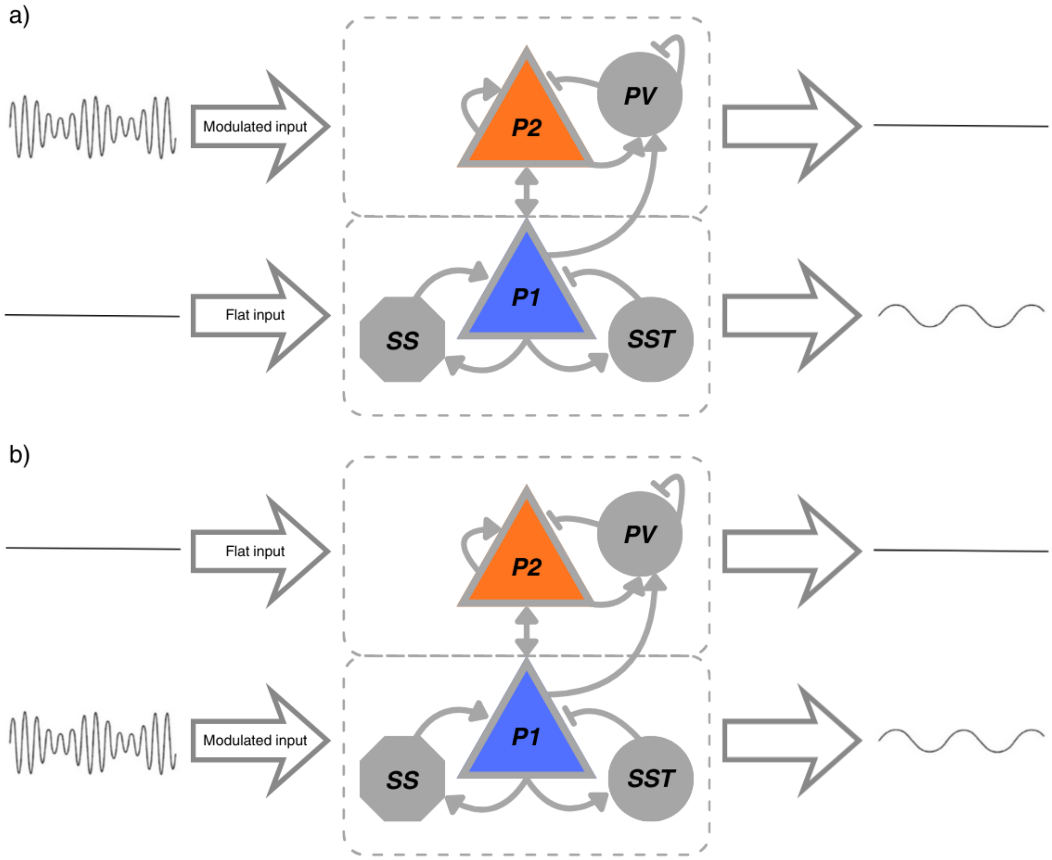

Figure 3: Stimulation setup in a LaNMM column. The column couples a superficial PING subcircuit (pyramidal P_2 + PV; gamma) and a deep JR subcircuit (pyramidal P_1 + SS + SST; alpha). *(a)* the AM input drives P_2 while P_1 receives a flat drive; *(b)* the AM input drives P_1 while P_2 is flat. Through the sigmoid nonlinearities and P_1–P_2 coupling, the slow envelope is transferred to the column: P_2 carries the modulated fast carrier, while P_1 exposes the recovered low-frequency envelope (right traces).

measure suited to the entrainment regime, where lock-in phase slips would otherwise complicate the read-out. Two sweeps probe the column: an amplitude–beat sweep over $(A, \Delta f)$ at fixed carrier (the Arnold tongue), and a carrier–beat sweep over $(f_c, \Delta f)$ at fixed A. This code is in `lanmm_arnold_tongues.py`, built on the LaNMM module `lanmmv11.py` of [9]. For the alpha resonance of Figs. 12, 13 we instead drive the deep P_1 loop directly and sweep its input against Δf, reading the lock-in (Eq. (14)) of v_{P_1} at Δf (self-contained `lanmm_resonance.py`).

Finally, we test the mechanism in the exact next-generation mean field (NMM2, §2.1), where the firing-rate nonlinearity is the derived MPR dynamics of Eq. (3) rather than a postulated sigmoid. We use a two-population excitatory–inhibitory (PING) motif [14, 15, 11]: each population $\alpha \in \{E, I\}$ is an MPR mean field (firing rate r_α, mean potential v_α) with membrane time τ_α, coupled through second-order synapses,

$$\begin{aligned}
\tau_E \dot r_E &= \tfrac{\Delta}{\pi \tau_E} + 2 r_E v_E, &\qquad \tau_E \dot v_E &= v_E^2 + \bar\eta - (\pi \tau_E r_E)^2 + \tau_E C\,(s_E - s_I) + \delta V(t),\\
\tau_I \dot r_I &= \tfrac{\Delta}{\pi \tau_I} + 2 r_I v_I, &\qquad \tau_I \dot v_I &= v_I^2 + \bar\eta - (\pi \tau_I r_I)^2 + \tau_I C\,(s_E - 2 s_I),
\end{aligned} \tag{16}$$

with second-order synaptic activations $s_E = \hat K_a[r_E]$ (excitatory, time constant τ_a) and $s_I = \hat K_g[r_I]$ (inhibitory, τ_g), each obeying the same critically-damped operator as the JR synapse, $\tau^2 \ddot s + 2\tau \dot s + s = r$; here C is the global synaptic gain and the weights $(A_{ee}, A_{ei}, A_{ie}, A_{ii}) = (1, 1, 1, 2)$ set the E and I rows. The common background drive $\bar\eta$—the same in both populations, with no separate input current—is the bifurcation parameter: raising it past a supercritical Hopf at $\bar\eta \approx 1$ opens a gamma limit cycle (the gamma counterpart of the JR alpha Hopf). The applied field enters the excitatory membrane current as $\delta V(t) = \lambda E(t)$ with the AM form of Eq. (1) (carrier $f_c = 300$ Hz, envelope

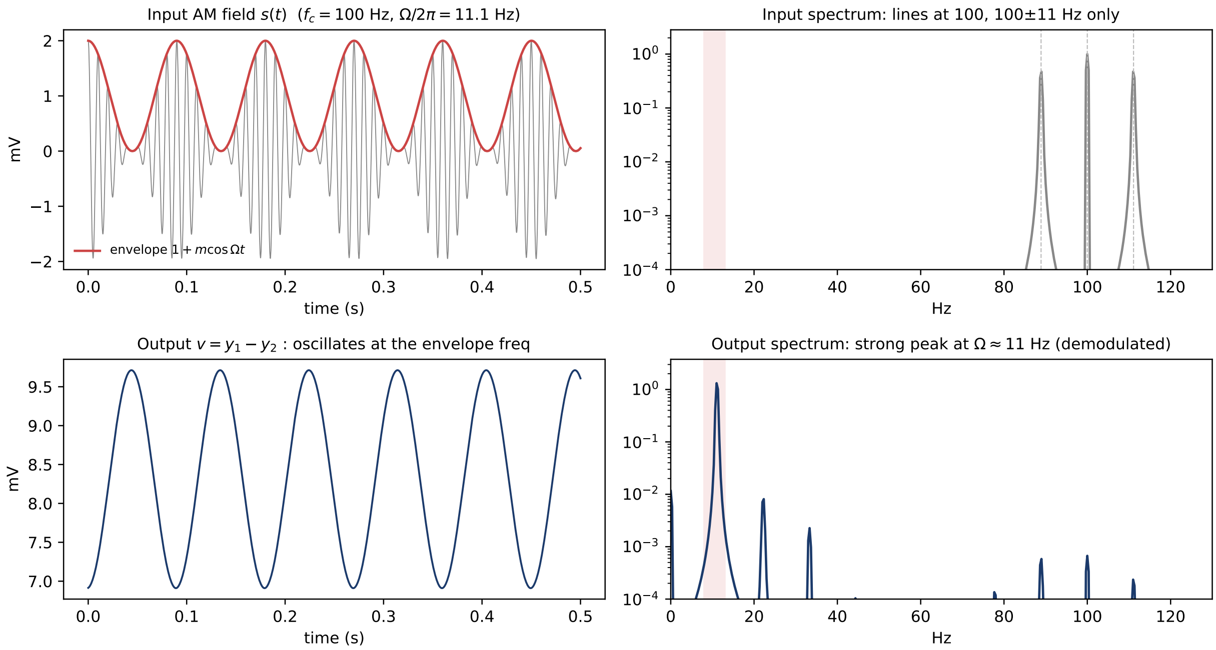

Figure 4: Demodulation in the Jansen–Rit column (proof of concept). *Top left:* the AM field $s(t)$ and its envelope. *Top right:* the input spectrum has lines only at f_c and $f_c \pm \Delta f$ (89, 100, 111 Hz); the alpha band (shaded) is empty. *Bottom left:* the output $v = y_1 - y_2$ oscillates at the envelope frequency. *Bottom right:* the output spectrum has a strong peak at $\Omega \approx 11$ Hz, synthesized by the sigmoid and absent from the input.

Δf swept through the gamma band), so the carrier is rectified by the exact quadratic v_E^2 term, with no postulated sigmoid. We integrate with fixed-step RK4 ($\Delta t = 0.05$ ms) and read out the lock-in (Eq. (14)) of r_E at Δf over the $(\Delta f, \bar{\eta})$ plane. Parameters (Rosetta Stone PING [11]): $\Delta = 1$, global coupling $C = 15$, $\tau_E = 15$, $\tau_I = 7.5$, $\tau_a = 10$, $\tau_g = 2.5$ ms, with $\bar{\eta}$ swept through the supercritical Hopf at $\bar{\eta} \approx 1$ ($f_0 \approx 55$ Hz). Code: `nmm2_ping.py` (self-contained).

4 Results

We present the evidence on three rungs of the modeling ladder: the single Jansen–Rit column (closed form, analyzable), the laminar LaNMM (two native resonances and genuine carrier dependence), and the exact next-generation mean field (NMM2, the demodulating nonlinearity derived rather than postulated). Each rung shows the same pair—square-law demodulation followed by near-Hopf amplification.

4.1 The Jansen–Rit sigmoid demodulates: an alpha line from a carrier-only input

The system is oriented by Fig. 2: a clean supercritical Hopf at $p \approx 315$ with $f_0 \approx 11.1$ Hz, and a sigmoid whose curvature peaks well off the inflection, so the closed-loop operating points sit where σ'' is large. The core result is Fig. 4: an input with no spectral power in the alpha band (lines only at f_c and $f_c \pm \Delta f$) produces a clean $\sim$11 Hz output oscillation locked to the envelope, with a strong alpha peak in the output spectrum that is entirely synthesized by the nonlinearity.

4.2 Near the Hopf, the demodulated envelope drives a sharp resonance

The resonance is quantified in Figs. 5 and 6. On the stable side ($p > 315$) the response is a Lorentzian centered at f_0 whose height grows monotonically as the Hopf is approached ($0.23 \to 0.43 \to 0.70$ mV), i.e. gain $\propto 1/\gamma$; on the limit-cycle side ($p < 315$) the demodulated drive entrains the autonomous alpha rhythm, with a peak near f_0 that deepens into the cycle. The two-

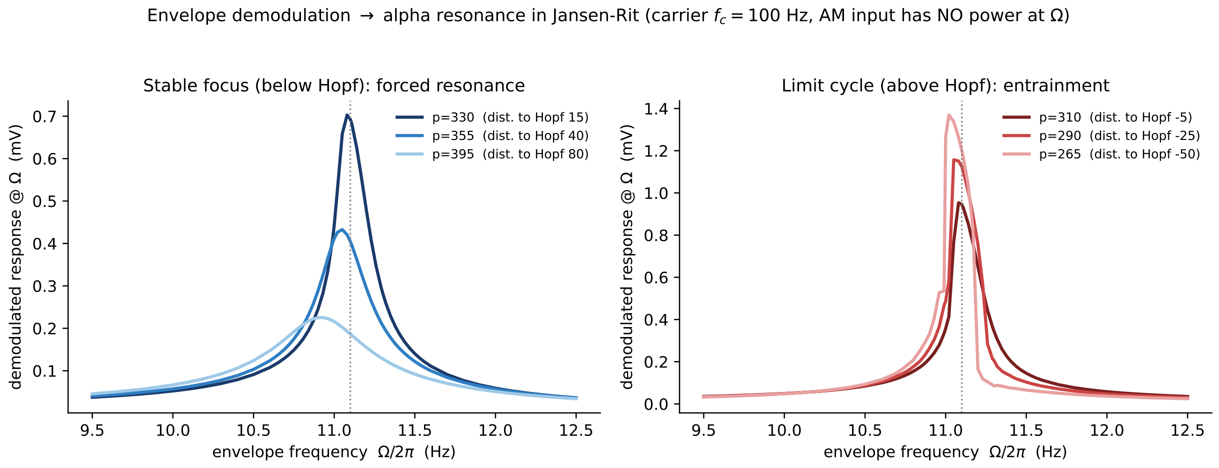

Figure 5: Resonance near vs. far from the Hopf (Jansen–Rit). Lock-in response at Ω vs. envelope frequency ($\varepsilon = 0.3$ mV). *Left (stable focus, $p > 315$):* forced resonance at $f_0 \approx 11.1$ Hz; the peak grows monotonically as $p \to 315$, i.e. gain $\propto 1/\gamma$. *Right (limit cycle, $p < 315$):* the demodulated drive entrains the autonomous alpha rhythm.

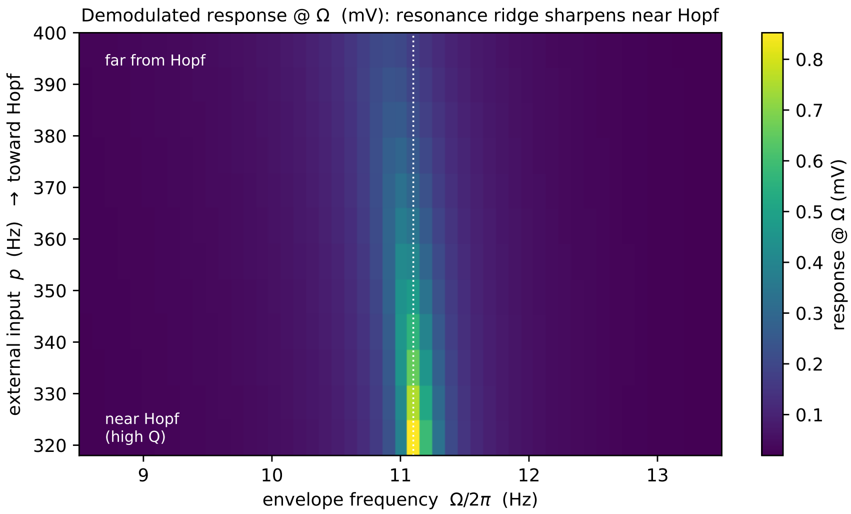

Figure 6: Resonance map (Jansen–Rit). Demodulated response A_Ω over the (envelope frequency, input p) plane. The ridge at $f_0 \approx 11.1$ Hz sharpens and intensifies as p approaches the Hopf from above, visualizing the $1/\gamma$ gain of Eq. (11).

dimensional map shows the same story as a ridge at f_0 that sharpens and intensifies toward the bifurcation.

4.3 Carrier independence and the kHz roll-off

At fixed post-coupling polarization the demodulated response is flat across the carrier band (Fig. 7, left): the open-loop detector reproduces Eq. (10) exactly, and the closed-loop JR near the Hopf adds $\sim$60$\times$ resonant gain without introducing any f_c dependence, deviating only as f_c approaches the natural band. As a function of the *applied* field, by contrast, the membrane low-pass makes the response roll off as $1/f_c^2$ above the corner (Fig. 8): passive somatic demodulation is suppressed four to six orders of magnitude at kHz, whereas a fast element ($\tau \approx 0.2$ ms) retains an appreciable response into the kHz band.

To pre-empt the objection that the demonstration runs only at 100 Hz, we drive the full recurrent column with a genuine kHz-carrier AM field, passing the field through an explicit membrane low-

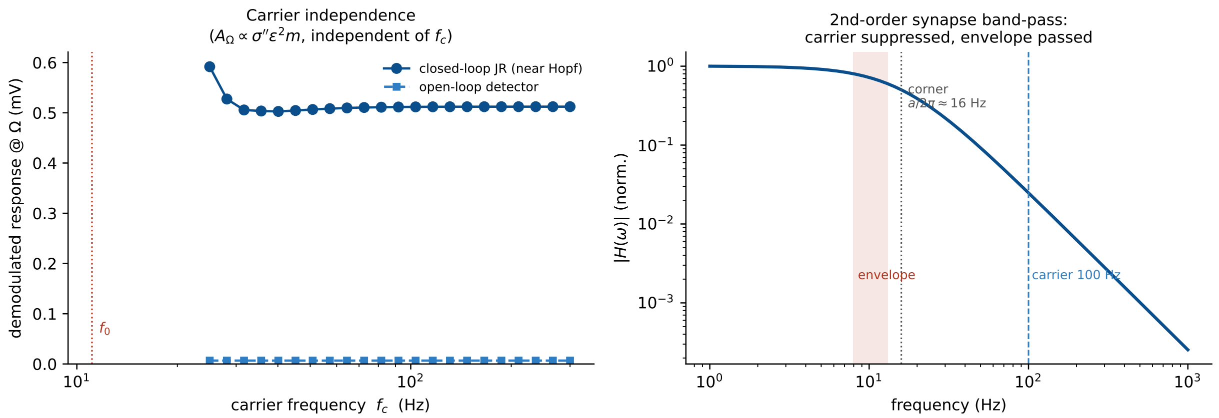

Figure 7: Carrier independence (Jansen–Rit single column). *Left:* the demodulated response is flat across carrier frequency $f_c = 25$–300 Hz, both for the open-loop detector (exactly, by Eq. (10)) and for the closed-loop JR near the Hopf (which adds $\sim 60\times$ resonant gain); deviations appear only as f_c approaches the natural band. *Right:* the second-order-synapse band-pass $|H(\omega)|$ passes the envelope (~ 0.7) and suppresses the 100 Hz carrier ($\sim 2.5\%$).

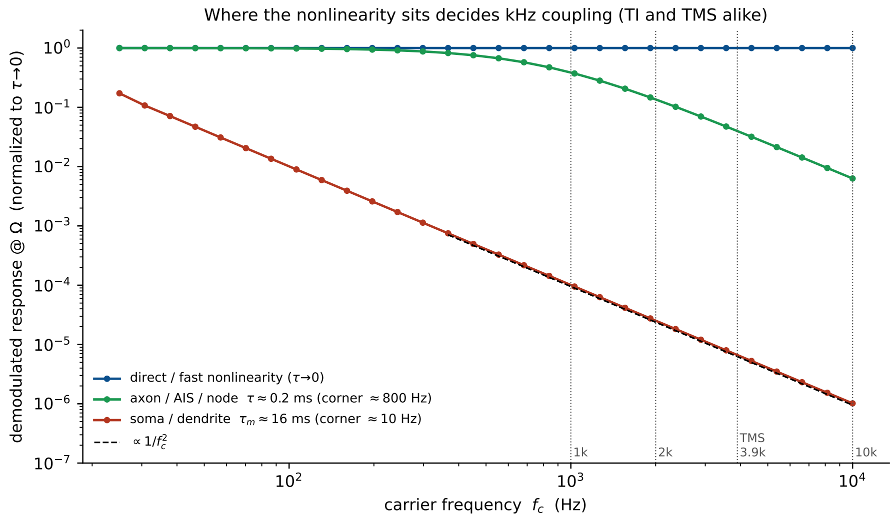

Figure 8: Carrier frequency in the kHz regime (Jansen–Rit, open-loop detector). Open-loop demodulated response (normalized to the $\tau \to 0$ value) vs. carrier frequency up to 10 kHz. With direct coupling ($\tau \to 0$) the response is flat—carrier-independent. A slow somatic membrane ($\tau_m \approx 16$ ms) rolls off as $1/f_c^2$ (dashed guide), killing kHz demodulation; a fast element (axon/AIS/node, $\tau \approx 0.2$ ms) retains an appreciable response into the kHz band (~ 0.14 at 2 kHz, ~ 0.04 at the TMS peak 3.9 kHz). The same membrane filter attenuates TMS at the soma, which is why neither TMS nor TI should be thought of as driving the soma at kHz.

pass (an extra first-order state, time constant τ) for each coupling route before it reaches the sigmoid (Fig. 9a). The column demodulates and resonates at the recovered alpha well into the kHz band when the nonlinearity sits at a fast element ($\tau \approx 0.2$ ms): the closed-loop A_Ω is carrier-independent for direct coupling, follows the fast element into the kHz, and collapses as $1/f_c^2$ for the slow soma ($\tau_m \approx 16$ ms), exactly as Eq. (13) predicts. A 2 kHz carrier through the fast element yields a clean alpha line synthesized by the column from an input with zero alpha power (Fig. 9b). That the bare unmodulated kHz carrier is itself weakly effective—the premise of TI's focality—is consistent with the limited sensitivity of hippocampal network oscillations to unmodulated kHz fields [22] (code: `khz_direct.py`).

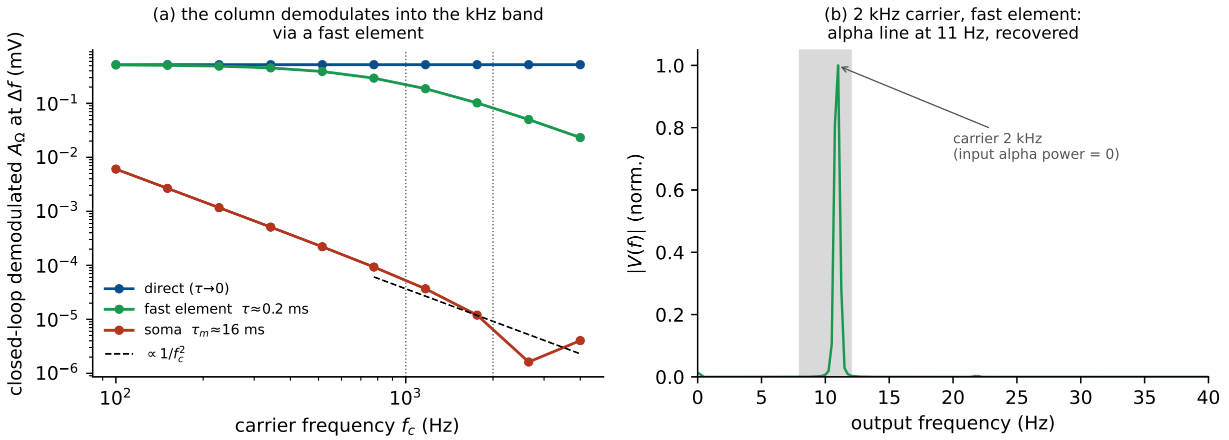

Figure 9: Demodulation at a genuine kHz carrier (Jansen–Rit, closed loop). The applied field passes through an explicit membrane filter before the sigmoid. *(a)* Closed-loop demodulated A_Ω at $\Delta f = f_0$ vs. carrier f_c for three coupling routes: direct ($\tau \to 0$, carrier-independent), fast element (axon/AIS/node, $\tau \approx 0.2$ ms; survives into the kHz band), and soma ($\tau_m \approx 16$ ms; collapses as $1/f_c^2$). *(b)* Output spectrum for a 2 kHz carrier delivered through the fast element: a clean alpha line at f_0, synthesized by the column although the input carries no alpha power.

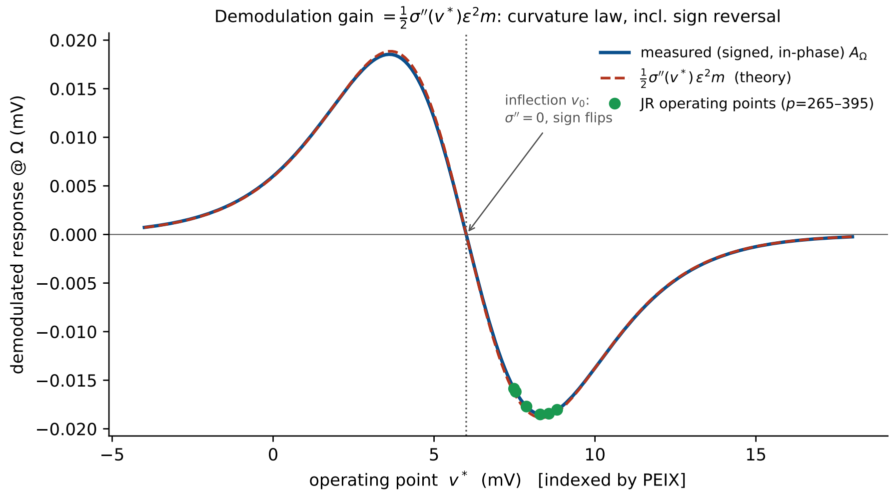

Figure 10: Operating-point law and the σ'' control (Jansen–Rit). The measured signed (in-phase) open-loop response tracks $\frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ across the whole range, vanishes at the inflection v_0, and reverses sign across it—a direct verification that demodulation is the sigmoid-curvature term of Eq. (10). The closed-loop JR operating points (dots) sit on the lobe where σ'' is large and negative.

4.4 The operating-point law and the σ'' control

Because $A_\Omega \propto \sigma''(v^*)$, the signed (in-phase) open-loop response tracks $\frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ across the whole operating range, vanishes at the inflection v_0, and reverses sign across it (Fig. 10)—a direct verification that demodulation is the sigmoid-curvature term of Eq. (10). The closed-loop JR operating points sit on the lobe where σ'' is large and negative.

4.5 The nonlinearity is the detector: square-law scaling and a linearization control

Fig. 11 closes the argument: the response scales as ε^2 at small drive (measured log–log slope 1.99 against the predicted 2), and replacing the field-receiving sigmoid by its linearization collapses the response by 712× (1.40→0.0020 mV): the nonlinearity is the detector.

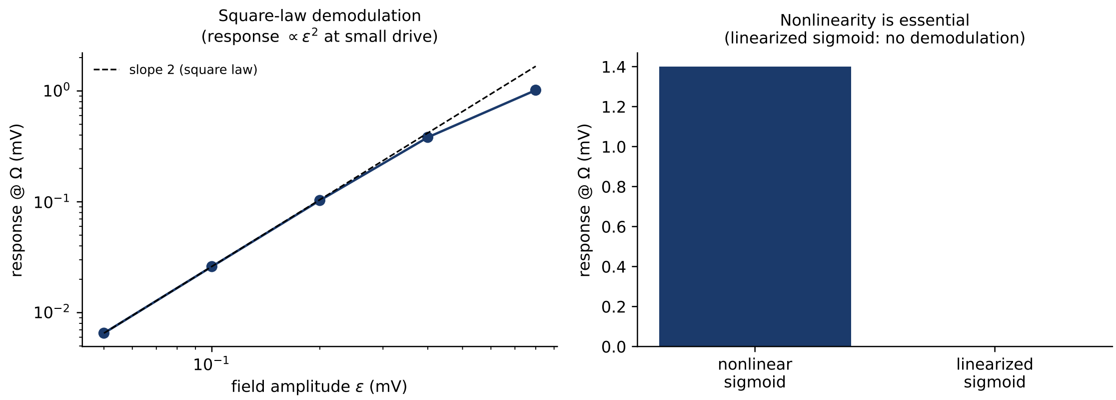

Figure 11: Verification (Jansen–Rit): square-law scaling and the linearization control. *Left:* response vs. field amplitude; small-drive log–log slope = 1.99 (theory 2). *Right:* linearizing the field-receiving sigmoid collapses the response 712× (1.40 → 0.0020 mV)—the nonlinearity is the demodulator.

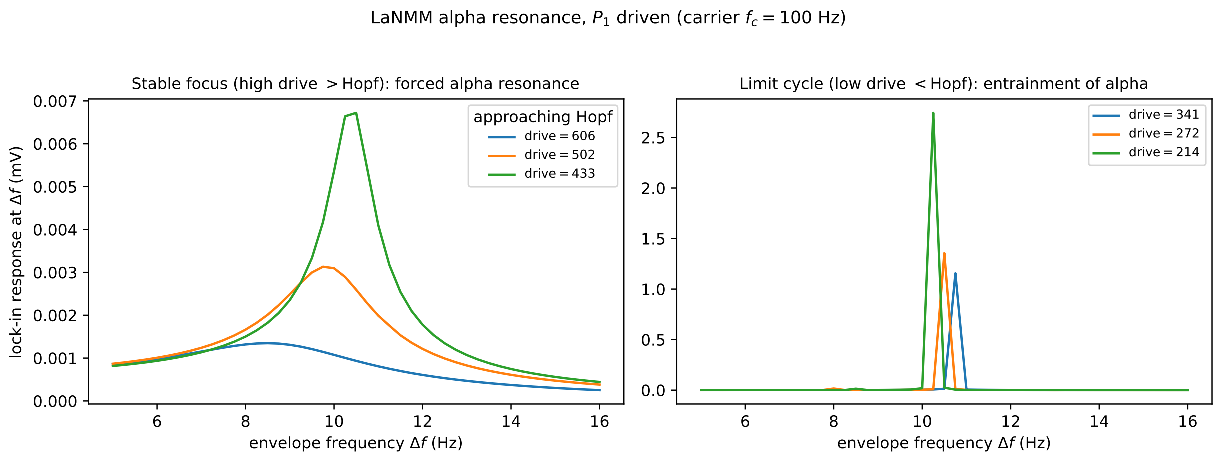

Figure 12: LaNMM alpha resonance (NMM1, laminar). Lock-in response of the deep pyramidal P_1 at the envelope frequency Δf, the P_1 (alpha) loop driven by an AM field (carrier 100 Hz) with its input as bifurcation parameter. *Left (stable focus, high drive):* forced alpha resonance at ∼10 Hz growing as the Hopf is approached. *Right (limit cycle, low drive):* entrainment of the autonomous alpha. Directly parallel to the JR single column (Fig. 5).

4.6 Two native resonances and nonlinear Arnold tongues in the LaNMM

The single JR column isolates and lets us analyze the mechanism; the laminar model (LaNMM, §2.1) confirms it in a cortical column with its *own* two resonances. Driving one population with the AM input of Eq. (15) and reading the alpha-band power of the deep (P_1, alpha) and superficial (P_2, gamma) pyramidal potentials (Methods, Fig. 3), we recover the same envelope resonance, now carrying the column's intrinsic rhythms.

Driving the deep P_1 (alpha) loop directly and sweeping the envelope frequency Δf reproduces the single-column picture in the laminar model (Figs. 12, 13). With the P_1 input as the bifurcation parameter, the stable-focus side (high drive) gives a forced alpha resonance at ∼10 Hz that grows as the alpha Hopf is approached, while the limit-cycle side (low drive) entrains the autonomous alpha—the same stable-focus/limit-cycle pair as the JR column (Fig. 5), now read out as the deep-pyramidal alpha lock-in. The heuristic laminar model thus shows the JR resonance *and*, in addition, the carrier-dependent Arnold tongues that follow.

Sweeping the modulation amplitude A and beat frequency Δf at a fixed carrier produces *nonlinear Arnold tongues*: alpha power in P_1 is maximal when Δf matches the deep alpha (∼10 Hz) and

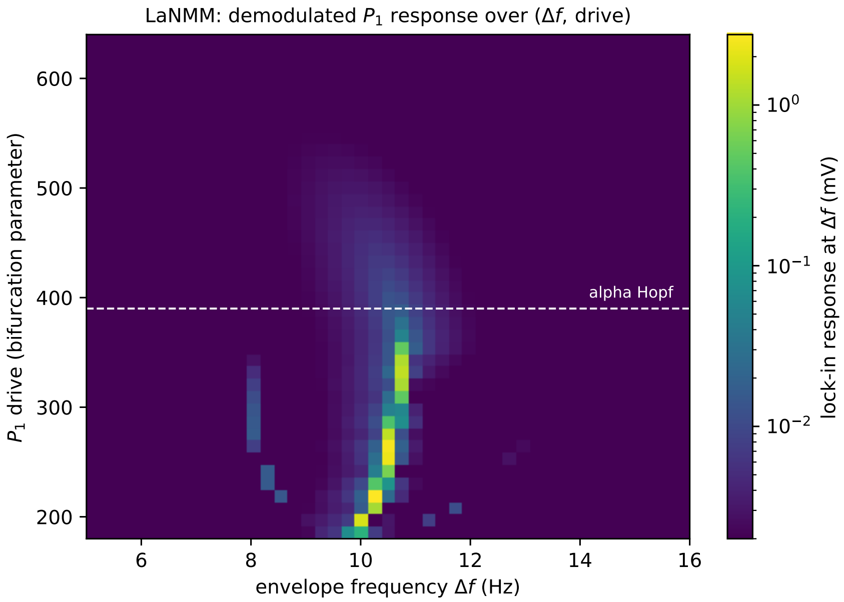

Figure 13: LaNMM alpha resonance map (NMM1, laminar). Demodulated P_1 response (log scale) over the (envelope frequency Δf, P_1 drive) plane: a ridge at the alpha natural frequency ($\sim$10 Hz) that intensifies through the supercritical Hopf (dashed line). The analog of the JR map (Fig. 6) and the NMM2 map (Fig. 17).

grows with A (Figs. 14a,b and 15a,b)—envelope resonance and phase locking of the slow generator to the recovered envelope, exactly as in the JR resonance curves but now in a biophysical two-band column.

The carrier dependence reproduces, and physically grounds, the two coupling limits of §2.2–§2.5. Sweeping carrier f_c against beat Δf: when the AM input is delivered to the *superficial* loop (P_2), strong P_1 demodulation requires a *double* resonance—f_c near the P_2 gamma ($\sim$40 Hz) *and* Δf near the P_1 alpha ($\sim$10 Hz) (Fig. 14c,d): the fast loop acts as a tuned "RF front end" that must itself be excited before it can pass a demodulated envelope to P_1. When the input is delivered *directly* to the slow generator (P_1), the response is broadband in f_c—a vertical ridge at $\Delta f \approx 10$ Hz across the whole carrier range (Fig. 15c), i.e. *carrier-independent*, just as Eq. (10) predicts for direct coupling. Real TI, in which the field reaches deep generators through faster elements, sits between these limits: a carrier matched to a fast (gamma) network resonance couples best, but the envelope response persists—weaker—off resonance. This both confirms the JR mechanism and refines it: the "inert carrier" of §2.3 is the direct-coupling idealization, while a band-limited intermediary makes the carrier choice matter, consistent with frequency-tagging/intermodulation paradigms [23].

Concretely the LaNMM predicts that TI efficacy should (i) peak when Δf matches a region's intrinsic alpha; (ii) widen and strengthen with modulation amplitude (the tongue); and (iii) for fields delivered via superficial fast circuits, be largest when the carrier is matched to a gamma network resonance—e.g. two nearby gamma carriers summing to a $\sim$40 Hz-centered AM field—while remaining present, weaker, for off-resonance carriers [24, 25].

4.7 Exact mean-field confirmation: the NMM2 PING gamma generator

The JR column and the LaNMM both use the heuristic firing-rate sigmoid (NMM1). As a third, independent test we move to the exact next-generation mean field (NMM2, Eq. (3)), where the demodulating nonlinearity is the *derived* quadratic v^2 term rather than a postulated curve. We drive the two-population E–I PING motif of Eq. (16) (Methods)—a gamma generator that loses stability through a supercritical Hopf (at $\bar{\eta} \approx 1$, $f_0 \sim 55$ Hz), the gamma counterpart of the JR

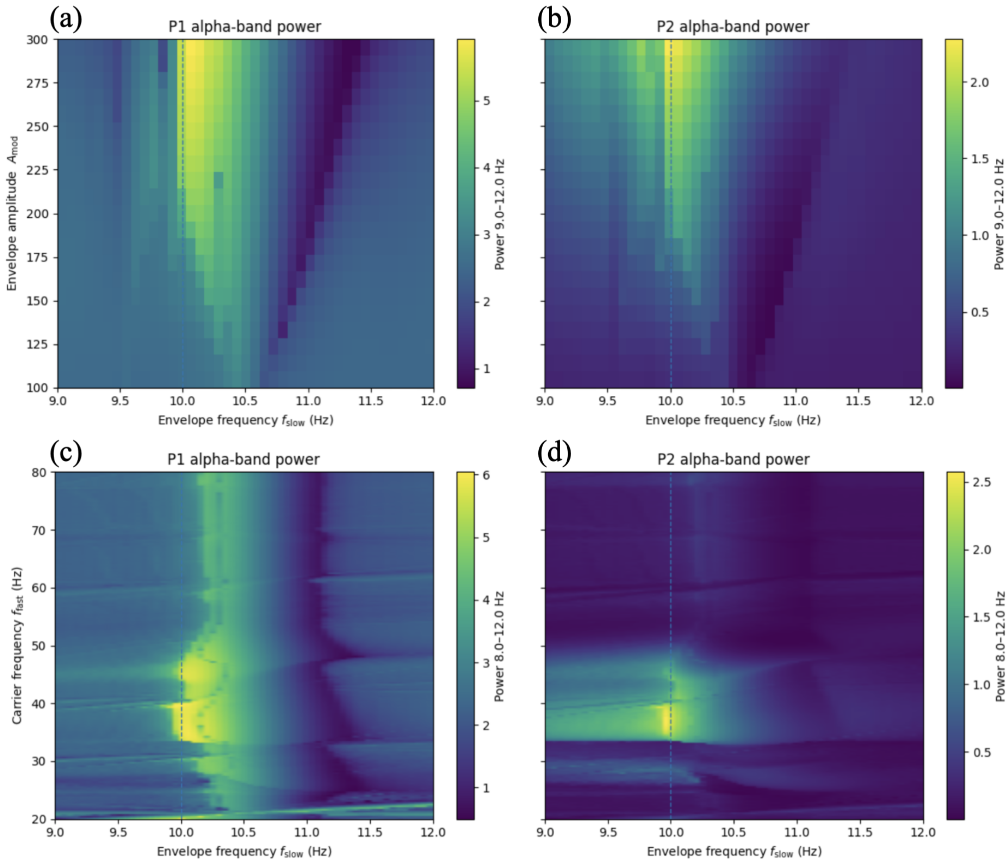

Figure 14: LaNMM Arnold tongues: AM input to the superficial P_2 (gamma) loop. *(a,b)* Alpha-band power of P_1 and P_2 vs. modulation amplitude A and beat frequency Δf (carrier 40 Hz): a tongue of elevated power around the P_1 alpha (∼10 Hz, dashed line) that widens with A. *(c,d)* Alpha power vs. carrier frequency and Δf at $A = 250$: P_1 demodulation requires *both* a carrier ≈40 Hz (P_2 gamma) and Δf ≈10 Hz (P_1 alpha)—a double resonance.

alpha Hopf—with the AM field entering the excitatory current, so the carrier is rectified by the exact v_E^2 term.

Driving with an AM field (carrier $f_c = 300$ Hz, envelope Δf swept through the gamma band) and locking in the excitatory rate r_E at Δf reproduces the whole picture (Figs. 16, 17). Below the Hopf the response is a forced resonance at the gamma frequency whose height grows as the bifurcation is approached—the $1/\gamma$ gain of Eq. (11); above it the demodulated drive entrains the autonomous gamma. The two-dimensional map traces a ridge at $\Delta f \approx f_0$ that intensifies through the Hopf. Unlike the static-sigmoid models, the resonant frequency itself *shifts with the operating point*—the ridge is diagonal, not vertical—a direct signature of the dynamic, state-dependent transfer of the exact theory [15]. That the same demodulation-and-near-Hopf-resonance mechanism appears with the nonlinearity *derived* from QIF spiking dynamics, rather than assumed, shows it is not an artifact of the heuristic sigmoid (code: `nmm2_ping.py`, self-contained).

4.8 What the mesoscale adds: coupling sets the gain at fixed detection

The resonance sweeps above drove the column toward the Hopf with the external input p, but p also shifts the operating point, entangling detection (σ'') with amplification. The sharper, mechanistic statement is that the *network coupling* sets the gain on its own. In the JR column we vary the global connectivity C—the coupling strength J—and, for each C, solve analytically for the drive $p(C)$ that *pins* the operating point v^*, so the detection gain $\sigma''(v^*)$ and the open-loop detector amplitude $A_{\text{open}} = \frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ are held exactly constant. As $C \to C^*$ (the Hopf, $C^* \approx 137$)

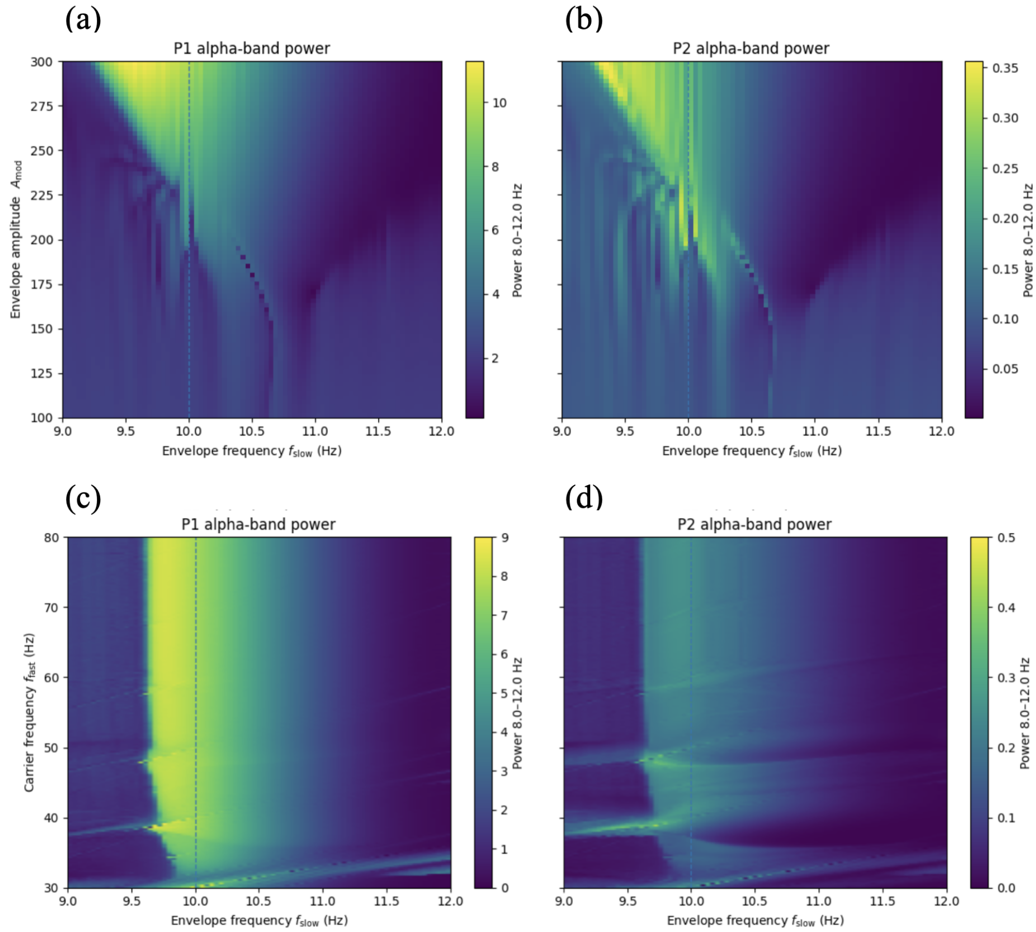

Figure 15: LaNMM Arnold tongues: AM input delivered directly to the deep P_1 (alpha) loop. *(a,b)* Alpha power of P_1 and P_2 vs. amplitude A and Δf. *(c,d)* Alpha power vs. carrier frequency and Δf: P_1 responds across a *broad* carrier band (a vertical ridge at $\Delta f \approx 10$ Hz)—carrier-independent, the direct-coupling limit of Eq. (10).

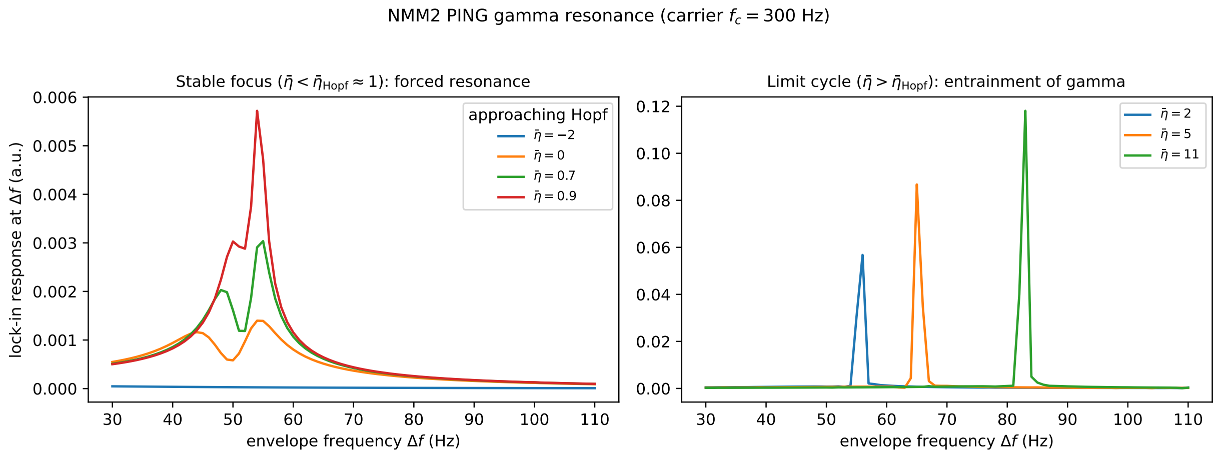

Figure 16: NMM2 PING gamma resonance (exact mean field). Lock-in response of the excitatory rate r_E at the envelope frequency Δf under an AM field (carrier 300 Hz). *Left (stable focus, $\bar{\eta} < \bar{\eta}_{\mathrm{Hopf}} \approx 1$):* forced resonance at the gamma frequency ($f_0 \approx 55$ Hz), growing as the Hopf is approached ($1/\gamma$ gain), with the resonant frequency rising with $\bar{\eta}$. *Right (limit cycle, $\bar{\eta} > \bar{\eta}_{\mathrm{Hopf}}$):* entrainment of the autonomous gamma. The input carries no power at Δf; the response is synthesized by the exact quadratic v^2 nonlinearity.

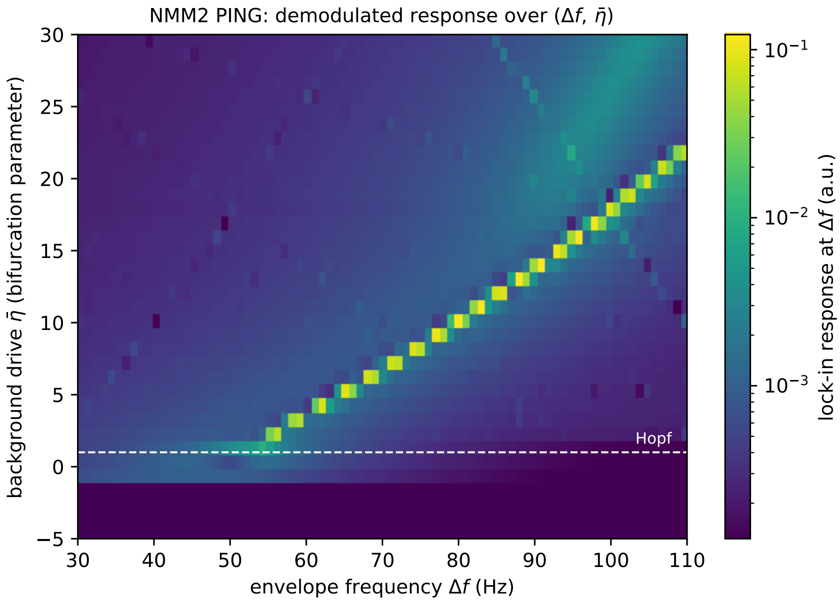

Figure 17: NMM2 PING resonance map (exact mean field). Demodulated response over the (envelope frequency Δf, background drive $\bar{\eta}$) plane (log color scale: entrainment far exceeds the stable-focus forced response). The ridge follows the gamma natural frequency $f_0(\bar{\eta})$ and intensifies through the supercritical Hopf (dashed line); its diagonal slope—f_0 rising with $\bar{\eta}$—reflects the state-dependent transfer of NMM2, absent in the static-sigmoid JR and LaNMM.

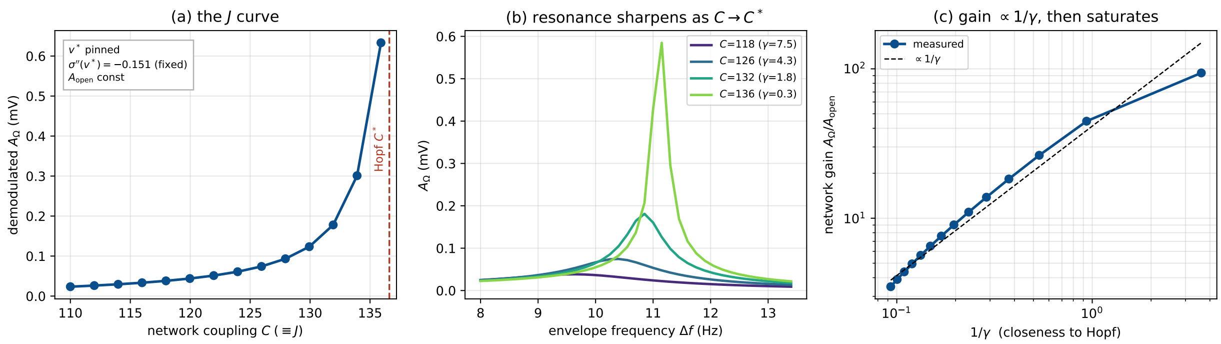

Figure 18: Coupling controls the amplification at fixed detection (Jansen–Rit). The operating point v^* is pinned by co-solving $p(C)$, so $\sigma''(v^*)$ and A_{open} are constant. *(a)* The demodulated A_Ω diverges as the coupling C $(= J)$ approaches the Hopf C^*. *(b)* The envelope resonance sharpens and grows as $C \to C^*$. *(c)* The network gain A_Ω/A_{open} follows $1/\gamma$ (dashed) and saturates near the bifurcation.

the damping $\gamma \to 0$, the demodulated response grows $\sim 27\times$ and the envelope resonance sharpens (Fig. 18a,b)—entirely a network effect, since detection is frozen and only γ changes. The recovered gain A_Ω/A_{open} tracks $1/\gamma$ before saturating near criticality (Fig. 18c); driving the same column to the Hopf through the input p rather than the coupling C yields the same gain–γ relation, so the amplification depends only on the distance to criticality, not on which parameter delivers it.

This becomes exact in the next-generation theory. In the NMM2 PING the coupling C is the *literal* mean inter-neuron (QIF) synaptic weight J, and the demodulating nonlinearity is the exact v_E^2 term, whose quadratic coefficient is a fixed constant—so detection needs no pinning. Sweeping J at fixed background drive $\bar{\eta}$ toward the gamma Hopf J^*, the demodulated peak of r_E grows as $1/\gamma$ and saturates at criticality (Fig. 19): the heuristic JR result, reproduced with the coupling *and* the rectifier both derived rather than assumed, and a direct counterpart of the coupling-controlled weak-field sensitivity of QIF networks [15]. The two columns realize one band-agnostic law in

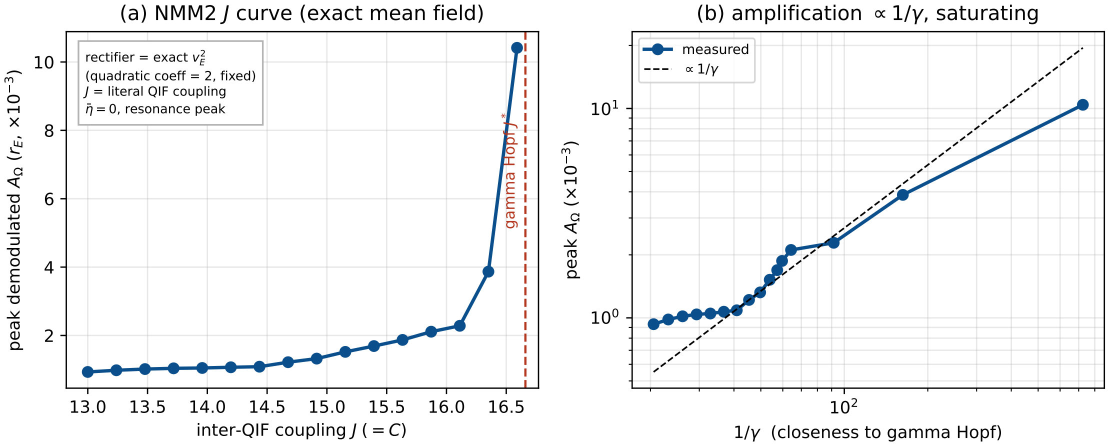

Figure 19: Coupling controls the amplification in the exact mean field (NMM2 PING, gamma). $J\,(=C)$ is the literal inter-QIF synaptic weight; the rectifier is the exact v_E^2 (fixed quadratic coefficient), so no operating-point pinning is needed. *(a)* The peak demodulated response of r_E diverges as $J \to J^*$ (gamma Hopf, $\bar{\eta} = 0$). *(b)* The gain grows as $1/\gamma$ (dashed), saturating near criticality—the JR law, now with coupling and rectifier both derived.

their two native bands—alpha (JR) and gamma (NMM2): the single-cell nonlinearity detects the envelope (coarse-grained into σ'', or the exact v^2), while the sharp, tunable gain is the network's—a resonant amplification set by proximity to the Hopf and adjustable through the coupling.

4.9 Timing, not rate: the resonance amplifies the AC envelope, not the mean rate

The near-Hopf gain is sharply frequency-selective, and this has a sharp experimental consequence. The demodulator feeds the network both a baseband (DC) term and a line at the beat Ω (Eq. (9)), but the transfer $G(\Omega)$ of Eq. (11) treats them very differently: at the resonance $G(\omega_0) \propto 1/(2\gamma\omega_0)$ diverges as the Hopf is approached, while at zero frequency $G(0) \propto 1/\omega_0^2$ is flat. Holding detection fixed (pinning v^*, exactly as in Fig. 18) and driving the column toward the Hopf through the coupling, the AC lock-in at the resonant beat $\Delta f = f_0$ grows $\sim$17×—tracking $1/\gamma$ before saturating near criticality—while the DC shift of the mean pyramidal firing rate stays flat at a few mHz, and even changes sign, throughout (Fig. 20). Near criticality the recovered envelope entrains population *timing* at Δf while leaving the mean *rate* essentially unchanged. This is the mesoscopic counterpart of the in-vivo observation that TI alters spike timing but not firing rate in the primate brain [6]: the network selectively amplifies the envelope-locked AC response, not the DC rate, so a near-flat mean rate is the expected signature even where TI is acting (code: `timing_not_rate.py`, `make_timing_fig.py`).

4.10 Falsifiable predictions and tests

The mechanism makes three sharp predictions, each with a clean experimental handle.

1. **Carrier independence at fixed polarization.** Once the post-coupling polarization $\varepsilon = \kappa(f_c)E_0$ is matched, the envelope response no longer depends on f_c, even though the *applied*-field threshold rises with f_c through the membrane roll-off (Eq. (13))—as single-neuron accounts also predict [4, 3]. *Test:* hold the in-zone polarization fixed (raising E_0 to compensate the measured $\kappa(f_c)$) while sweeping f_c; the network signature is a flat envelope response.
2. **Operating-point (PEIX) control and sign flip.** Because $A_\Omega \propto \sigma''(v^*)$, the response scales with the curvature at the operating point and *reverses sign* across the inflection, vanishing at v_0 where the gain σ' (PEIX, Eq. (12)) peaks. *Test:* shift v^* with tonic drive, neuromodulation, or task state and read out the predicted scaling and sign reversal of the TI response.

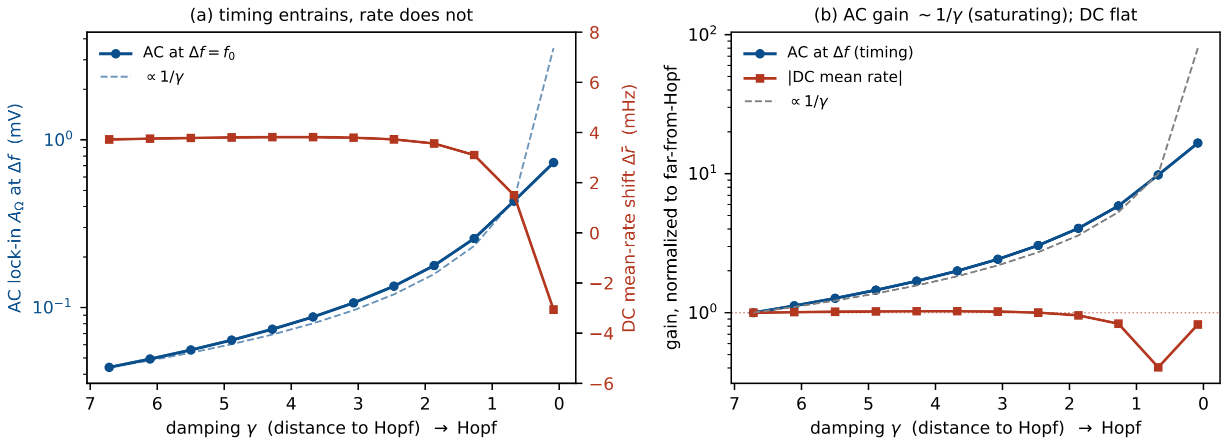

Figure 20: Timing, not rate (Jansen–Rit, detection pinned). As the column is driven toward the Hopf through the coupling at fixed $\sigma''(v^*)$, *(a)* the AC lock-in at the resonant beat $\Delta f = f_0$ (blue) is resonantly amplified and tracks $1/\gamma$ (dashed), while the DC shift of the mean firing rate (red) stays flat at a few mHz. *(b)* Normalized to the far-from-Hopf value (log scale): the AC gain follows $1/\gamma$ and saturates near criticality, whereas the DC gain stays $\sim$1. The envelope entrains timing, not rate—the population analog of the spike-timing-without-rate finding of [6].

3. **Frequency selectivity and state-dependence.** The response is largest when Δf matches a region's resonant rhythm and is amplified as $1/\gamma$ near a bifurcation, so manipulations that move the operating point or the distance to the Hopf—psychedelics, PV-interneuron dysfunction, attention, arousal—should scale, and can abolish or sign-reverse, TI efficacy [9]. *Test:* sweep Δf through a region's intrinsic band across brain states and track the resonant peak and its width.

5 Discussion

We have given a mesoscopic, population-level account of TI demodulation in which the firing-rate sigmoid is a square-law detector and proximity to a Hopf bifurcation turns the recovered envelope into a sharp, frequency-selective response. This is complementary to—not in competition with—the single-neuron ion-channel rectification account [3]: both are rectify-then-filter demodulation, but ours needs no specific channel, lives at the scale of the EEG-generating population, and inherits the brain's own resonances, so a region's tuning is set by its intrinsic dynamics just as a radio is tuned by its tank. Because each mass is $\sim 10^4$ neurons, demodulation is a *population* read-out by the sigmoid—a single-neuron nonlinearity coarse-grained over the ensemble—while the amplification (the near-Hopf resonance) is the emergent *network* property; the single cell, beyond contributing to the sigmoid's shape, need only supply a fast nonlinearity that samples the carrier.

This factorization—a fast single-cell nonlinearity samples the carrier, after which the population reads out the envelope and the network resonantly amplifies it—identifies where the carrier-sampling nonlinearity may live. The TI literature points to three best-supported, non-exclusive loci. (i) *Axonal Na^+ channels* at nodes and the AIS: fast voltage-gated Na^+ rectifies because activation outpaces inactivation [3, 26], Na^+-inactivation conduction block passes the slow envelope while removing the carrier [3, 4], and persistent Na^+—a fast subthreshold rectifier tied to axonal resonance—could even set an optimal Δf. (ii) *Presynaptic terminals*, which polarize $\sim$2–10× more than somata [27, 28, 29] and convert depolarization into transmitter release through the steeply nonlinear Ca^{2+}–release relation. (iii) *Threshold spiking*, where near threshold the spike probability is a smooth nonlinearity and a weak bias is rectified through stochastic resonance [20, 17]. In model cortical neurons TI produces block, onset and phasic responses depending on geometry [26], and in vivo it alters spike timing rather than rate in the primate brain [6]—consistent with the timing-not-rate signature of our resonance (§4.9). The mesoscopic mechanism is agnostic to which of these does the sampling.

The same fast-element logic explains TMS, and underscores that high-frequency drive is not somatic. TMS pulses carry kHz spectral content (peak ∼3.9 kHz), which a $\tau_m \approx 16$ ms soma attenuates by ∼50 dB; TMS—like TI—therefore cannot be driving the soma at kHz, and works instead through fast excitable elements whose chronaxie (∼0.15–0.25 ms) sets an effective corner near ∼800 Hz [30, 31]. That the bare unmodulated kHz carrier is itself weakly effective on network oscillations [22] is the complement of the same fact: the carrier matters only where a fast nonlinearity rectifies it.

The mechanism's falsifiable predictions—carrier independence at fixed polarization, the PEIX-controlled sign flip, and frequency-selective state-dependence—are consolidated with their tests in §4.10; they could be read out in the same models already applied to Alzheimer's disease and psychedelics [9]. The two-band LaNMM results (§4.6) bear this out: its JR (alpha) and PING (gamma) subcircuits give the same sigmoid demodulation two native resonances, with the beat frequency selecting the band; this embeds TI demodulation in the HAM read-out hierarchy [10], where envelope extraction is the inverse of the amplitude-modulation encoding used for prediction-error evaluation.

Several limitations should be kept in view. The field enters the population sigmoid directly; we treat the field-to-nonlinearity coupling $\kappa(f_c)$ phenomenologically (Eq. (8)) rather than modeling channel kinetics, and ε should be read as the post-coupling polarization, not the applied field (§2.2). The population sigmoid is likewise a coarse-grained stand-in for distributed single-cell nonlinearities. JR alpha near this Hopf is high-Q, so the stable-side peaks are narrow and would be broadened by noise and population heterogeneity. The 100 Hz carrier is below TI's kHz range, but this is a testable regime in its own right (§2.5) and the argument depends only on a timescale separation. Finally, the LaNMM realization here still treats a single, deterministic, noiseless column with a phenomenological field coupling; coupled noisy columns and an explicit, realistic $\kappa(f_c)$ remain natural next steps.

6 Conclusion

A population sigmoid and a second-order-synapse network near a Hopf bifurcation suffice to demodulate a temporal-interference field and resonate at the recovered envelope frequency. The sigmoid's curvature is inherited from single-neuron nonlinearity, coarse-grained over the $\sim 10^4$ neurons of the mass; what is emergent is that demodulation is read out at the *population* level—needing no specific channel—and that the resonant amplification is a *network* property. A fast single-cell nonlinearity (Na^+ at nodes/AIS, persistent Na^+, presynaptic terminals) need only sample the carrier. The column then behaves as a tuned AM receiver—a nonlinear detector followed by a resonant tank—so that *wherever a fast-enough nonlinearity sees the carrier, the population sigmoid demodulates and the network amplifies.*

The amplification half is not automatic: the gain scales as $1/\gamma$ with the distance γ to the Hopf, so a large, sharply tuned response appears only near criticality (high excitability); far from the bifurcation the envelope is recovered but barely amplified. The amplification is therefore not only state- but *coupling*-tunable: holding detection fixed, the demodulated gain grows toward $1/\gamma$ as the network is driven to criticality by its own connectivity (§4.8), in the heuristic JR column and, with the inter-neuron coupling and the rectifier both derived, in the exact NMM2 mean field. Both halves are thus strongly **state-dependent**—detection through the curvature $\sigma''(v^*)$ (which even changes sign across the inflection), amplification through γ—and both are set by neuromodulation, attention, arousal and pathology (PEIX). Under this mechanism TI efficacy is a property of *brain state* as much as of the stimulus: largest in near-critical regimes, and tunable—even abolished or sign-reversed—by shifting the operating point. This gives a concrete, carrier-independent, state-dependent mesoscopic mechanism for TI, complementary to single-neuron rectification, tying TI to cross-frequency coupling and hierarchical amplitude modulation.

References

[1] Nir Grossman, David Bono, Nina Dedic, Suhasa B. Kodandaramaiah, Andrii Rudenko, Ho-Jun Suk, Antonino M. Cassara, Esra Neufeld, Niels Kuster, Li-Huei Tsai, Alvaro Pascual-Leone, and Edward S. Boyden. Noninvasive deep brain stimulation via temporally interfering electric fields. *Cell*, 169(6):1029–1041.e16, 2017.

[2] Paul J. Nahin. *The Science of Radio: with MATLAB and Electronics Workbench Demonstrations*. Springer, 2nd edition, 2001.

[3] Ehsan Mirzakhalili, Beatrice Barra, Marco Capogrosso, and Scott F. Lempka. Biophysics of temporal interference stimulation. *Cell Systems*, 11(6):557–572.e5, 2020.

[4] Aleksandar Opančar, Petra Ondráčková, David Samuel Rose, Jan Trajlinek, Vedran Đerek, and Eric Daniel Głowacki. The same biophysical mechanism is involved in both temporal interference and direct kHz stimulation of peripheral nerves. *Nature Communications*, 16:9006, 2025.

[5] Ryan B. Budde, Michael T. Williams, and Pedro P. Irazoqui. Temporal interference current stimulation in peripheral nerves is not driven by envelope extraction. *Journal of Neural Engineering*, 20(2):026032, 2023.

[6] Pedro G. Vieira, Matthew R. Krause, and Christopher C. Pack. Temporal interference stimulation disrupts spike timing in the primate brain. *Nature Communications*, 15:4558, 2024.

[7] Sara Caldas-Martinez, Chaitanya Goswami, Mats Forssell, Jiaming Cao, Alison L. Barth, and Pulkit Grover. Cell-specific effects of temporal interference stimulation on cortical function. *Communications Biology*, 7:1076, 2024.

[8] Hossein Soroushi, Sara Abbasi, Yun Du, Ning Ning, and Yuming Lei. Temporal interference stimulation: mechanisms, optimization, validation, and clinical prospects—a comprehensive review. *WIREs Computational Statistics*, 17(3):e70031, 2025.

[9] Giulio Ruffini, Edmundo Lopez-Sola, Raul Palma, Roser Sanchez-Todo, Jakub Vohryzek, Francesca Castaldo, and Karl Friston. Cross-frequency coupling as a neural substrate for prediction error evaluation: a laminar neural mass modeling approach. *bioRxiv*, 2025. Preprint 2025.03.19.644090.

[10] Giulio Ruffini. Neural encoding through hierarchical amplitude modulation. *bioRxiv*, 2025. Preprint 2025.11.03.686310.

[11] Francesca Castaldo, Raul de Palma Aristides, Pau Clusella, Jordi Garcia-Ojalvo, and Giulio Ruffini. Rosetta Stone of neural mass models. *Physics Reports*, 2025. accepted; arXiv:2512.10982.

[12] Ben H. Jansen and Vincent G. Rit. Electroencephalogram and visual evoked potential generation in a mathematical model of coupled cortical columns. *Biological Cybernetics*, 73(4):357–366, 1995.

[13] Ernest Montbrió, Diego Pazó, and Alex Roxin. Macroscopic description for networks of spiking neurons. *Physical Review X*, 5(2):021028, 2015.

[14] Giulio Ruffini. Analysis and extension of exact mean-field theory with dynamic synaptic currents. *bioRxiv*, 2021. Preprint 2021.09.01.458563 (NMM2).

[15] Pau Clusella, Elif Köksal-Ersöz, Jordi Garcia-Ojalvo, and Giulio Ruffini. Comparison between an exact and a heuristic neural mass model with second-order synapses. *Biological Cybernetics*, 117(1-2):5–19, 2023.

[16] Roser Sanchez-Todo, André M. Bastos, Edmundo Lopez-Sola, et al. A physical neural mass model framework for the analysis of oscillatory generators from laminar electrophysiological recordings. *NeuroImage*, 270:119938, 2023.

[17] Davide Reato, Asif Rahman, Marom Bikson, and Lucas C. Parra. Low-intensity electrical stimulation affects network dynamics by modulating population rate and spike timing. *Journal of Neuroscience*, 30(45):15067–15079, 2010.

[18] Daniella Terney, Leila Chaieb, Vera Moliadze, Andrea Antal, and Walter Paulus. Increasing human brain excitability by transcranial high-frequency random noise stimulation. *Journal of Neuroscience*, 28(52):14147–14155, 2008.

[19] Onno van der Groen and Nicole Wenderoth. Transcranial random noise stimulation of visual cortex: stochastic resonance enhances central mechanisms of perception. *Journal of Neuroscience*, 36(19):5289–5298, 2016.

[20] Anli Liu, Mihály Vöröslakos, Greg Kronberg, Simon Henin, Matthew R. Krause, Yu Huang, Alexander Opitz, Ashesh Mehta, Christopher C. Pack, Bart Krekelberg, Antal Berényi, and Lucas C. Parra. Immediate neurophysiological effects of transcranial electrical stimulation. *Nature Communications*, 9:5092, 2018.

[21] François Grimbert and Olivier Faugeras. Bifurcation analysis of Jansen's neural mass model. *Neural Computation*, 18(12):3052–3068, 2006.

[22] Zeinab Esmaeilpour, Mark Jackson, Greg Kronberg, Tianhe Zhang, Rosana Esteller, Brad Hershey, and Marom Bikson. Limited sensitivity of hippocampal synaptic function or network oscillations to unmodulated kilohertz electric fields. *eNeuro*, 7(6):ENEURO.0368–20.2020, 2020.

[23] Nathan Gordon, Jakob Hohwy, Matthew J. Davidson, Jeroen J. A. van Boxtel, and Naotsugu Tsuchiya. From intermodulation components to visual perception and cognition—a review. *NeuroImage*, 199:480–494, 2019.

[24] Zeinab Esmaeilpour, Greg Kronberg, Davide Reato, Lucas C. Parra, and Marom Bikson. Temporal interference stimulation targets deep brain regions by modulating neural oscillations. *Brain Stimulation*, 14(1):55–65, 2021.

[25] Ines R. Violante et al. Non-invasive temporal interference electrical stimulation of the human hippocampus. *Nature Neuroscience*, 26, 2023.

[26] Boshuo Wang, Aman S. Aberra, Warren M. Grill, and Angel V. Peterchev. Responses of model cortical neurons to temporal interference stimulation and related transcranial alternating current stimulation modalities. *Journal of Neural Engineering*, 19(6):066047, 2023.

[27] Asif Rahman, Davide Reato, Mattia Arlotti, Fernando Gasca, Abhishek Datta, Lucas C. Parra, and Marom Bikson. Cellular effects of acute direct current stimulation: somatic and synaptic terminal effects. *The Journal of Physiology*, 591(10):2563–2578, 2013.

[28] Aman S. Aberra, Ruochen Wang, Warren M. Grill, and Angel V. Peterchev. Multi-scale model of axonal and dendritic polarization by transcranial direct current stimulation in realistic head geometry. *Brain Stimulation*, 16(6):1776–1791, 2023.

[29] Darpan Chakraborty, Dennis Q. Truong, Marom Bikson, and Hanoch Kaphzan. Neuromodulation of axon terminals. *Cerebral Cortex*, 28(8):2786–2794, 2018.

[30] Anthony T. Barker, C. W. Garnham, and Ian L. Freeston. Magnetic nerve stimulation: the effect of waveform on efficiency, determination of neural membrane time constants and the measurement of stimulator output. *Electroencephalography and Clinical Neurophysiology Supplement*, 43:227–237, 1991.

[31] Angel V. Peterchev, Stefan M. Goetz, Gregory G. Westin, Bruce Luber, and Sarah H. Lisanby. Pulse width dependence of motor threshold and input–output curve characterized with controllable pulse parameter transcranial magnetic stimulation. *Clinical Neurophysiology*, 124(7):1364–1372, 2013.